How Many Factors Are Real on the Moscow Exchange?

A Pre-Specified Multiple-Testing Study of the Russian Factor Zoo

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October 2026

Abstract

Do the return anomalies documented in developed markets exist on the Moscow Exchange once the number of tests is taken into account? I test implementations of 25 price-, trading- and payout-based predictors on MOEX-listed equities (holding months 2012:01–2026:09), with data free of survivorship bias for shares but covering depositary receipts only in part, under a protocol fixed before estimation, and evaluate each under a ladder of seven significance criteria. Seven characteristics clear $|t| > 1.96$ in the predicted direction; five survive the Romano–Wolf stepdown, whose bootstrap critical value (3.39) exceeds both the Harvey–Liu–Zhu hurdle and the normal-theory Bonferroni bound: with fifteen years of monthly data, autocorrelation-robust t-statistics are more dispersed than the normal distribution that normal-theory corrections assume. Four survivors are closely related measures of intermediate-horizon momentum—nearness to the 52-week high, six- and twelve-month and residual momentum—and are largely redundant rather than four independent discoveries; the fifth, a high-volume premium, is fragile. The implemented low-risk, liquidity, size, skewness, seasonality and dividend-yield signals do not survive, but premia with annual Sharpe ratios below about one would often go undetected in a sample of this length. The one-month reversal is reversed: among liquid shares last month's winners keep winning ($t = -4.96$ against the reversal hypothesis). Momentum remains significant net of estimated spread costs, with short legs restricted to index shares and under bounds for unobserved delisting losses, but its capacity in equal-weighted form is small. After the 2022 exclusion of foreign investors no premium changes significantly once the 25 differences are corrected for multiplicity, yet the confidence intervals admit large changes in either direction.

JEL: G12, G14, G15, C12. **Keywords:** factor zoo, multiple testing, Romano–Wolf, replication, momentum, Russian stock market, MOEX.

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Аннотация

Существуют ли на Московской бирже аномалии доходности, найденные на развитых рынках, если учесть число проверяемых гипотез? В работе проверяются реализации 25 предикторов доходности, основанных на ценах, торговой активности и дивидендах, на выборке бумаг Мосбиржи (месяцы удержания 2012:01–2026:09), которая свободна от ошибки выжившего в части акций, но лишь частично охватывает депозитарные расписки, по протоколу, зафиксированному до оценивания; каждый оценивается по лестнице из семи критериев значимости. Порог $|t| > 1{,}96$ в предсказанном направлении проходят семь характеристик; пошаговую процедуру Романо–Вольфа выдерживают пять, причём её бутстреп-критическое значение (3,39) выше и порога Харви–Лю–Чжу, и нормального порога Бонферрони: на пятнадцатилетней месячной выборке t-статистики с поправкой на автокорреляцию распределены шире нормального закона. Четыре выживших сигнала — меры ценового моментума на горизонте 6–12 месяцев (близость к 52-недельному максимуму, моментум за 6 и 12 месяцев, остаточный моментум), которые во многом дублируют друг друга и не являются четырьмя независимыми открытиями; пятый, премия за аномальный объём торгов, неустойчив. Реализованные сигналы низкого риска, ликвидности, размера, асимметрии, сезонности и дивидендной доходности поправку не выдерживают, однако премии с годовым коэффициентом Шарпа ниже примерно единицы на выборке такой длины часто остаются незамеченными. Краткосрочный реверс имеет обратный знак: среди ликвидных акций победители прошлого месяца продолжают расти. Моментум остаётся значимым после оценённых спредовых издержек, при коротких позициях только в акциях индекса и при консервативных допущениях о потерях при делистинге, но его ёмкость в равновзвешенной форме мала. После ухода нерезидентов в 2022 г. ни одна премия не изменилась значимо с учётом поправки на множественность, однако доверительные интервалы допускают и крупные изменения в обе стороны.

1 Introduction

Fifty years of empirical asset pricing have produced hundreds of variables that predict the cross-section of stock returns—a "zoo" of factors in the words of Cochrane (2011). When so many hypotheses are tested on overlapping data, the conventional $|t| > 1.96$ threshold admits many false discoveries. Harvey et al. (2016) argue that a new factor should clear $t > 3$; Harvey and Liu (2020) make explicit the trade-off between false and missed discoveries. Large replication exercises disagree on how serious the problem is. Hou et al. (2020) find that 65% of 452 anomalies fail to clear $|t| = 1.96$ once microcaps are neutralised, whereas Jensen et al. (2023) conclude, with a hierarchical Bayesian model and data for 93 countries, that most factors replicate; Chen and Zimmermann (2022) reach a similarly benign verdict, and Chordia et al. (2020) quantify how much of the zoo multiple-testing control removes. Almost all of this evidence comes from the United States or from global panels in which any single emerging market carries little weight.

The Russian market has its own, smaller literature on return anomalies, and it reaches mixed conclusions. Teplova and Mikova (2014) find momentum profits only for short formation and holding periods and contrarian profits at horizons of six months and longer; Abramov et al. (2019) report significant size and liquidity factors over 1997–2017 with weakening after 2011; Ovechkin and Boldyreva (2022) find significant market and momentum premia but insignificant size and value premia over 2011–2020; Gurov (2023) finds the expected-illiquidity premium insignificant in most specifications. Each of these results is, in effect, a single test on its own sample, with its own definitions and portfolio rules, and none is adjusted for the number of hypotheses that the Russian and international literatures have examined. A reader cannot tell which of the reported anomalies would survive a common protocol and an honest correction for multiplicity.

Russia is also an unusually informative laboratory. Its cross-section is narrow and concentrated: the 100th most liquid share traded a median of only about 26 million roubles a month in 2012 and between 10 and 12 million in 2014–2016 (Table 1). Trading is dominated by individuals, who accounted for 79% of share trading volume in 2023 (Moscow Exchange, 2024). And the market went through a sharp, exogenous change in its investor base: equity trading was suspended at the end of February 2022, transactions with investors from "unfriendly" jurisdictions required special permission (President of the Russian Federation, 2022a), and when trading resumed for 33 index shares on 24 March 2022 (Bank of Russia, 2022c) the market had in effect been closed to foreign capital. If anomalies reflect limits to arbitrage, they should be stronger in such a market, and the 2022 break allows a before-and-after comparison within one market.

This paper tests 25 return predictors on all shares traded on the MOEX shares market, with data that are free of survivorship bias for shares but cover depositary receipts only in part (Section 3.2), under a protocol that was written, hashed and committed to version control before any characteristic was computed. The characteristics are those that can be built from prices, trading activity and dividends—momentum, reversal, seasonality, low risk, skewness, size, liquidity and payout. Each is implemented as closely to its source as daily market data and a common portfolio protocol allow and is signed by the direction the source reports (for two liquidity measures, by the paper that relates the measure to returns); Appendix Table 12 lists where the implementation departs from the source, for example CAPM rather than Fama–French residuals for residual momentum and idiosyncratic volatility, or daily rather than intraday or expected skewness. The tests therefore concern these implementations rather than exact replications of the original definitions. Each long–short portfolio is evaluated under a ladder of seven criteria: the

conventional $t > 1.96$, the $t > 3$ hurdle of Harvey et al. (2016), the Holm family-wise correction, the Benjamini–Hochberg and Benjamini–Yekutieli false-discovery-rate procedures, the Romano and Wolf (2005) stepdown test with a stationary bootstrap, and the Deflated Sharpe Ratio of Bailey and López de Prado (2014). The family-level probability of backtest overfitting (Bailey et al., 2017), the effective number of independent tests and an empirical-Bayes model in the spirit of Jensen et al. (2023) complete the picture. Every variant in a pre-specified robustness grid is run through the whole ladder.

The results are easy to state. Seven of the 25 characteristics clear $t > 1.96$ in the predicted direction; five survive the Romano–Wolf test. The Romano–Wolf critical value, 3.39, is larger than the Harvey–Liu–Zhu hurdle and even than the normal-theory Bonferroni bound of 3.09. The reason is not fat-tailed returns. Dependence among the factors lowers the bar, but by less than an effective number of about 15 independent tests would suggest; with 177 monthly observations the Newey–West standard error is itself estimated with noise, so the t-statistics are more dispersed than the standard normal; and serial dependence in the returns adds to that dispersion. Normal-based corrections such as Holm and Benjamini–Hochberg are therefore anticonservative in samples of this length: in simulations under the global null, Holm commits a family-wise error in 8.0–9.8% of samples and the $t > 3$ rule in 9.7–11.5%, against 5.8–7.3% for the bootstrap test, which is itself mildly liberal. Twelve-to-seven-month momentum passes Holm and Benjamini–Hochberg but fails the bootstrap test ($p_{\mathrm{RW}} = 0.058$).

Four of the five survivors are forms of intermediate-horizon price momentum: nearness to the 52-week high (George and Hwang, 2004), with an annualised Sharpe ratio of 1.44, six- and twelve-month momentum (Jegadeesh and Titman, 1993) and residual momentum (Blitz et al., 2011). They are not four independent discoveries: their returns are strongly correlated, and six- and twelve-month momentum are spanned by the 52-week high and residual momentum, so the robust evidence is better described as one momentum family; the data do not pin down how many distinct effects that family contains. The fifth survivor is the high-volume return premium of Gervais et al. (2001). Residual momentum survives the Romano–Wolf test in every full-sample specification; the high-volume premium does not survive a one-session implementation lag, the 50 most liquid names, price-only returns or a size-calibrated version of the test. None of the implemented low-risk, size, liquidity, skewness, seasonality, long-term reversal or dividend-yield signals survives, even at the conventional threshold. The one-month reversal of Jegadeesh (1990) is reversed: among liquid shares, last month's winners outperform last month's losers by 1.96% a month ($t = -4.96$ against the reversal hypothesis). The continuation survives a one-session implementation lag, so it is unlikely to be a bid–ask artefact. In exploratory analysis it is a property of the liquid segment as a whole, while weak signs of the usual reversal appear only in the lowest turnover tercile of the broad cross-section, in line with the short-term momentum of Medhat and Schmeling (2022).

The momentum survivors remain significant net of spread-based trading costs, with break-even one-way costs of 148–252 basis points, and they survive bounds for the unobserved losses of delisted shares. They also stay significant when short positions are confined to the most liquid shares or to index constituents and during the 2022 short-sale ban, and a borrow fee would have to exceed 15–32% a year to eliminate their net premia. Their capacity is small, however: under a square-root price-impact rule the net premia of the equal-weighted portfolios vanish at between 41 and 167 million roubles per leg over the full sample, and at higher but still modest sizes after 2022. The exclusion of foreign investors in 2022 did not change any individual premium significantly once the 25 differences are corrected for multiplicity,

and no significant change is detected in the average anomaly premium (0.49% versus 0.70% a month). Absence of evidence is not evidence of no change: the 95% confidence interval for the change in the average premium runs from -0.23 to 0.64 percentage points a month, and the before-and-after comparison cannot separate the exit of foreign capital from the other shocks of 2022. Several liquidity-related premia switch from negative to positive after 2022, but the evidence is not strong enough to survive correction. Finally, the power analysis shows what fifteen years of an emerging market can and cannot reveal: the Romano–Wolf test detects a factor with an annual Sharpe ratio of 0.5 with probability 0.13 and one with a Sharpe ratio of 1.0 with probability 0.72. The survivors have Sharpe ratios between 1.02 and 1.44; many anomalies documented in the United States are much weaker, so their failure to replicate in Russia is weak evidence that they are absent.

The paper contributes in four ways. It provides the first multiple-testing study of the factor zoo on the Russian market, with a common protocol applied to share data free of survivorship bias. It shows that with the fifteen-year samples typical of emerging markets the dependence-robust bootstrap bar is higher, not lower, than the familiar $t > 3$ rule, because autocorrelation-robust t-statistics are more dispersed than the normal distribution, so that normal-theory corrections overstate significance. It documents robust short-term continuation among liquid Russian shares. And it uses the 2022 change in the investor base to ask whether anomaly premia respond to the exit of foreign arbitrage capital. The pre-analysis plan, all deviations from it and the code are part of the replication package.

Section 2 reviews the literature. Section 3 describes the market and the data. Section 4 sets out the pre-specified design. Section 5 reports the results, Section 6 discusses them, and Section 7 concludes.

2 Related literature

Multiple testing and the factor zoo. The concern that data mining manufactures return predictability goes back at least to Lo and MacKinlay (1990). White (2000) proposed the bootstrap reality check for the best of many strategies, Hansen (2005) refined it, and Romano and Wolf (2005) turned it into a stepdown procedure that controls the family-wise error rate while exploiting the dependence among test statistics; Romano and Wolf (2016) give the adjusted p-values used here. False-discovery-rate control (Benjamini and Hochberg, 1995) is valid under independence or positive dependence, and the Benjamini and Yekutieli (2001) version under arbitrary dependence at a cost in power. In finance, Harvey et al. (2016) catalogue hundreds of published factors and derive a hurdle near $t = 3$ for new ones; Harvey and Liu (2020) weigh false against missed discoveries. Hou et al. (2020) replicate 452 anomalies and find that most fail once microcaps are handled with NYSE breakpoints and value weights, while Jensen et al. (2023) use a hierarchical Bayesian model to argue that most factors replicate and that correlated factors should not be treated as independent tests. Chen and Zimmermann (2022) provide open-source replications; Chordia et al. (2020) show how many strategies mined from accounting data look significant by chance; Feng et al. (2020) and Giglio et al. (2021) develop tests for new factors and for thousands of alphas. McLean and Pontiff (2016) document that anomaly returns fall out of sample and after publication; Jacobs and Müller (2020) find that this decline is concentrated in the United States. Bailey and López de Prado (2014) correct the Sharpe ratio of the best of N trials for selection and non-normality, and Bailey et al. (2017) estimate the probability that the in-sample winner underperforms out of sample. This paper

applies all of these tools to one market on one protocol and reports them side by side.

Anomalies in emerging markets. Rouwenhorst (1999) documents momentum, size and value effects in emerging markets. Griffin et al. (2003) show that momentum profits are widespread internationally and are not explained by business-cycle risk. Bekaert et al. (2007) use the proportion of zero daily returns to show that liquidity is priced in emerging markets. Cakici et al. (2013) study 18 emerging markets over 1990–2011 and find a value effect in all of them and a momentum effect in all regions except Eastern Europe. Hanauer and Lauterbach (2019), using 28 emerging markets, find cash-flow-to-price, gross profitability, equity issuance and momentum to be pervasive and no positive relation between risk and return. These studies test a few factors at a time and, with the exception of the Bayesian evidence in Jensen et al. (2023), do not adjust for the number of tests; they also pool many countries, so a single market's idiosyncrasies are averaged away.

The cross-section of Russian stock returns. Early work focused on aggregate returns: Goriaev and Zabotkin (2006) analyse the risks of the market's first decade, and Anatolyev (2008) documents highly unstable relations between Russian index returns and their determinants. Teplova and Shutova (2011) find that the unconditional CAPM explains little of the cross-section of liquid Russian stocks and that downside-risk measures perform better. Teplova and Mikova (2014) report that the best momentum strategy on 134 stocks uses three-month windows and that contrarian strategies dominate at horizons of six months or more. Kinnunen and Martikainen (2017) find that idiosyncratic risk is priced at the industry level with a sign that flips after 2008. Abramov et al. (2019) build a five-factor model with significant market, size and liquidity factors over 1997–2017; Abramov et al. (2025) report that most of 15 long-only factor portfolios outperformed the MOEX total-return index over 2007–2024. Teplova and Gurov (2022) find that the effect of implicit trading costs on MOEX returns over 2010–2020 is concentrated in small capitalisations and significant mainly in time-series tests, and Gurov (2023) finds the expected-illiquidity premium insignificant in most specifications. Ovechkin and Boldyreva (2022) find significant market and momentum premia and insignificant size and value premia over 2011–2020, and Nazarova and Leshchev (2023) find momentum profitable after costs among highly liquid shares in 2019–2021. None of these studies corrects for multiple testing across factors, and their samples end before or shortly after the 2022 break.

3 Institutional background and data

3.1 The Russian equity market, 2012–2026

The Moscow Exchange (MOEX) is the only significant venue for Russian shares. Over the sample the main board moved from same-day (T0) to T+2 settlement, completed on 2 September 2013, and to T+1 on 31 July 2023 (Moscow Exchange, 2023). The settlement regime matters for the ex-dividend date, which precedes the dividend record date by the settlement lag (Section 3.3). On 28 February 2022 equity trading was suspended; transactions in which residents transfer securities to persons from jurisdictions designated as unfriendly became subject to special permission (President of the Russian Federation, 2022a); trading resumed for 33 shares of the MOEX Russia index on 24 March 2022, with short sales banned (Bank

of Russia, 2022c), and for all shares on 28 March 2022 (Moscow Exchange, 2022). The short-sale ban, in force from 24 February, was lifted on 1 June 2022 (Bank of Russia, 2022a). Payments to such foreign holders were subsequently directed to special type-C accounts (President of the Russian Federation, 2022b), and individuals became the dominant investor group (Moscow Exchange, 2024). The market is concentrated in a few large state-related and resource firms, and dividends are large and highly seasonal, which makes accurate dividend adjustment essential.

3.2 Data sources

All market data originate from the Moscow Exchange's Information and Statistical Server (ISS). Because the ISS endpoints were not reachable from the analysis environment, the data are taken from public mirrors that store raw ISS responses (Appendix B). Daily prices, volumes and rouble trading values for every share ever traded on the MOEX shares market come from the collection of Andreev (2026), which includes delisted shares. Its ticker dictionary lists 1034 securities (SECIDs), 758 of which were delisted before the end of the sample; daily price histories exist for 1012. Prices are stitched across trading boards, with the primary board taking precedence and the odd-lot board used only when nothing else traded. The same source provides a register of splits and reverse splits, dividends reconciled across several sources and checked against MOEX's own register-closing dates, and the MCFTRR total-return index, which is used as the market return. Historical numbers of issued shares come from the quarterly MOEX index calculation bases from December 2012 onwards (quantviews, 2026), and current share counts from the MOEX list of securities of May 2026. The risk-free rate is the Bank of Russia key rate (the refinancing rate before September 2013). The data contain ordinary and preferred shares, including shares of foreign issuers traded on the shares market. Depositary receipts are covered only in part. The source contains the receipt histories of six foreign issuers whose receipts were later converted into Russian shares under the same ticker (TCS Group, Ozon, VK, Etalon, MD Medical Group and United Medical Group), but not those of receipts that were delisted or replaced by shares with a new ticker. Eight such receipts were constituents of the MOEX Russia index at some point—those of Ros Agro, X5, Fix Price, Globaltrans, HeadHunter, Lenta and QIWI, and Rusal's Russian depositary receipts until 2015—and up to four index constituents at a time are therefore missing; the Russian successors of such issuers enter as new securities. The share segment of the sample is survivorship-free, but the receipt segment is not. The pre-analysis plan assumed that the source contained no receipts; Appendix A records the discrepancy, and Section 5.5 drops the six receipt histories. There are no accounting data, so the value, profitability, investment and accrual families of anomalies cannot be studied.

3.3 Returns, corporate actions and screens

Prices before a split are multiplied by the ratio of old to new shares, volumes are divided by it, and dividend amounts with a record date before the split are rescaled in the same way; 14 securities have recorded splits. Dividends are gross of tax. All dividend sources but one store the record date, and the ex-date is located on the share's own trading days: the session after the record date under T0, the session before it under T+2, and the record date itself under T+1. The exception, derived from Yahoo Finance, stores the ex-date, which is used directly; it supplies 25% of the payments used (15% of those in U100 holding months), and

its dates are checked in Appendix B. Of the dividend payments in the data, 2624 are used; 262 whose implied ex-date falls outside the record month or an adjacent month (the share did not trade around the record date), 121 that precede the first price and 30 paid in foreign currency are dropped. Russian issuers often pay a final and an interim dividend with the same record date; such payments are summed. A legal change affects the timing of payouts: from 1 January 2014 the dividend record date must fall 10 to 20 days after the decision to pay (Russian Federation, 2013), which moved most record dates from April–May, before the annual meeting, to June–July. As a result, in May and June 2014 the 2013 payouts had already left the trailing 12-month window used for dividend yield while the 2014 payouts had not yet arrived, and the number of U100 shares with a trailing dividend falls to 20 and 32 in those two months, against about 60 otherwise. The daily total return is $r_d = (P_d + D_d)/P_{d^-} - 1$, where d^- is the share's previous trading day. A daily return enters daily statistics only if the share also traded in the previous session and the gap between the two sessions is at most 14 calendar days, which removes the month-long 2022 suspension from volatility, beta and spread estimates. A single print that jumps by more than $+100\%$ or -60% and reverts to within 20% of the previous price is deleted; 18 prints are removed. The monthly return is the change in the total-return index between the last trades of consecutive months. A share that stops trading during a month earns the return to its last trade; a share held in a portfolio that does not trade at all in the holding month earns zero. These rules record no loss after the last trade: the final payoff of a delisted share—a buyout or exchange price, a write-off, or capital locked in an account—is not observed. In U100, 45 shares stop trading for good while in the universe: for 43 the last trade falls inside a holding month (their average return to the last trade is -13%), and for two it precedes the holding month. In 10 cases the share is still in U100 at the formation date after its last trade and is then held as cash for a month. Another three position-months fall in suspensions after which trading resumed. Section 5.5 bounds the effect of the terminal events, charging each of them once.

To validate the construction, I build a value-weighted portfolio of the constituents of the MOEX index bases with point-in-time share counts and compare it with MCFTRR. Over the 165 months from 2013 the monthly correlation is 0.988, and the portfolio earns 0.37 percentage points a year more than the index—the expected order of magnitude, since MCFTRR is net of dividend withholding tax and uses free-float weights with caps; the replication also omits the index constituents that are missing from the source (Section 3.2).

3.4 Universes and sample

At the end of each month t a share is *active* if it traded on at least $\min(10, \lceil n_t/2 \rceil)$ of the month's n_t sessions, its last trade lies within the final five sessions of the month, it traded in each of the past 12 months, and it is not marked as delisted. The cap at half the month's sessions binds in six formation months with 18 or fewer sessions and matters materially only in March 2022, which had six. When ISS records a change of ticker, the predecessor is used before the change and the successor afterwards; redomiciled securities are treated as new securities. The baseline universe U100 contains the 100 most liquid active shares by the median monthly trading value over months $t-11$ to t, a relative cut that keeps the breadth constant while the rouble value of trading grows by an order of magnitude. U50 contains the 50 most liquid, and UALL all active shares.

Holding months run from January 2012 to September 2026 ($T = 177$). On average 232 shares are

active; 423 different shares are active at some point, and 164 of them no longer trade, under the same or a new ticker, at the end of the sample. Of the 261 shares that ever enter U100, 90 (34%) are gone by September 2026, so a sample built from currently listed shares would miss about a third of the baseline universe. Table 1 and Figure 1 describe the sample. The liquidity floor of U100 falls from about 26 million roubles a month in 2012 to 10–12 million in 2014–2016 and rises to between 500 and 800 million in 2024–2026, as retail inflows and higher nominal turnover lifted trading across the board.

Market capitalisation, needed for size, turnover and value weights, is available point-in-time only for index constituents. For other shares I use the nearest index-base share count of the same security and otherwise the current share count; shares with neither have no capitalisation. Coverage of U100 rises from 71% in 2012 to 100% from 2024. Because the fallback counts contain look-ahead, a robustness variant uses point-in-time counts only.

Table 1: Sample description

Year	Months	N_{ALL}	U100 floor (RUB m/month)	Median cap (RUB bn)	Cap coverage (%)	Payers in U100	MCFTRR (%)	r_f (%)
2012	12	222	26.0	62.8	71	51	8.4	8.1
2013	12	222	19.6	64.6	71	58	5.9	7.6
2014	12	204	11.7	59.6	75	53	−2.3	7.3
2015	12	213	10.5	69.2	80	61	31.5	13.0
2016	12	240	12.3	98.0	81	60	32.0	10.6
2017	12	240	25.2	131.4	82	61	−0.9	9.2
2018	12	232	23.7	132.7	85	67	18.2	7.4
2019	12	240	20.2	150.9	84	69	37.1	7.4
2020	12	240	31.2	111.5	87	65	13.9	5.1
2021	12	241	52.8	151.7	90	69	20.9	5.6
2022	12	237	80.9	109.6	92	64	−38.0	11.0
2023	12	232	261.7	133.7	97	51	52.5	9.6
2024	12	227	801.0	124.1	100	54	0.5	17.4
2025	12	239	516.7	103.9	100	59	2.0	19.3
2026	9	245	508.0	98.2	100	59	−12.2	10.9

The sample consists of holding months 2012:01–2026:09; each row aggregates the formation months whose holding month falls in the year. N_{ALL}: average number of active shares (UALL). U100 floor: median across months of the trailing 12-month median monthly trading value of the least liquid share in U100. Median cap: median market capitalisation of U100 shares with a share count. Cap coverage: share of U100 with a capitalisation. Payers: average number of U100 shares that paid a dividend in the previous 12 months. MCFTRR: MOEX Russia total-return index (net of dividend tax), compounded over the holding months. r_f: compounded key rate. 2026 covers January–September.

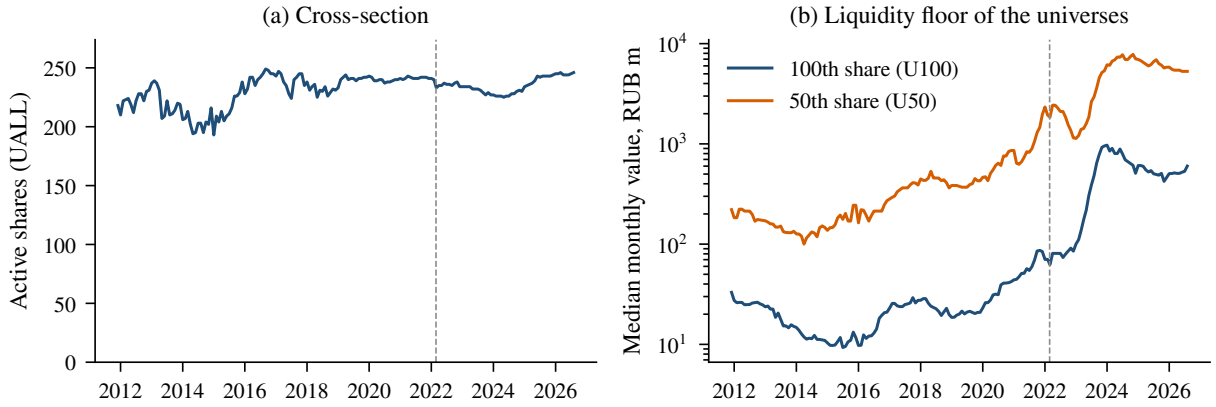


Figure 1: The cross-section of MOEX shares. Panel (a): number of active shares (UALL) at each formation month. Panel (b): trailing 12-month median monthly trading value of the least liquid share in U100 and U50 (log scale, millions of roubles). The dashed line marks the February 2022 suspension.

4 Pre-specified research design

4.1 The pre-analysis plan

The design was written down before estimation in a pre-analysis plan that fixes the sample, universes, the 25 characteristics with their windows, minimum data requirements and predicted signs, the portfolio rules, the test statistics, the ladder of corrections, the robustness grid, the hypotheses and a list of commitments. Before the plan was frozen, only diagnostics that do not condition on any characteristic had been examined: counts of shares, sessions and dividends; the unconditional distribution of next-month returns in U100; the market-replication check above; and the coverage of share counts. The plan was frozen on 10 October 2026 at 19:41 UTC; its SHA-256 hash and those of the configuration and data-construction code were recorded and committed to a version-controlled repository before the first characteristic was computed. This is an internal pre-specification rather than a filing with an external registry, and it does not erase the author's knowledge of the published literature; its purpose is to make every analytic choice visible and fixed in advance. Appendix A lists the deviations: a few implementation details not spelled out in the plan, a bug fix found by an independent code audit after the results were seen, a more efficient design of the power simulation, and several exploratory analyses added after seeing the results, which are labelled as such wherever they appear.

4.2 Characteristics

Table 2 lists the 25 characteristics. They fall into seven clusters: momentum (12-1, 6-1 and 12-7 month momentum, residual momentum and nearness to the 52-week high), reversal (one-month and long-term), seasonality, low risk (Dimson beta, idiosyncratic and total volatility, maximum daily return), skewness and downside risk (downside beta, total, idiosyncratic and co-skewness), size and liquidity (size, Amihud illiquidity, trading value, turnover, the zero-return proportion, the Corwin–Schultz spread, the variability of trading value and abnormal trading volume) and payout (dividend yield). Each is computed at the last session of month t from data available at that time; the exception is the fallback share counts behind size and turnover, which contain look-ahead and have a point-in-time variant (Section 3.4). Perturbation tests confirm that no characteristic, universe membership or cost estimate changes when the price and trading

data after t are altered (Appendix B). The predicted sign is the direction reported in the source paper (for the zero-return proportion and the spread, whose source papers propose measures, in the paper that relates the measure to returns).

The implementations follow the sources where daily market data allow and adapt them where they do not, and all portfolios follow the common protocol below whatever the holding rule of the source. Appendix Table 12 lists the differences. The substantive ones are four. Residual momentum and idiosyncratic volatility use CAPM rather than Fama–French three-factor residuals, because without book equity there is no value factor; Section 5.5 re-estimates both with a two-factor model that adds a size factor built from the sample. Total skewness is measured from daily returns over a year rather than as the intraday realised skewness of Amaya et al. (2015), and idiosyncratic skewness is historical rather than the model-based expected skewness of Boyer et al. (2010). The beta portfolio is a low-minus-high quintile spread on Dimson (1979) betas rather than the beta-neutral, leveraged portfolio of Frazzini and Pedersen (2014), so its CAPM alpha is the closer test of their prediction. Several windows and horizons also differ (coskewness from daily rather than monthly returns; abnormal volume over a month rather than a day or a week; long-term reversal, seasonality and the variability of trading value over shorter histories). A failure to survive is therefore a statement about the implemented signal, not necessarily about the original anomaly.

Table 2: The 25 characteristics

ID	Cluster	Definition (at the last session of month t)	Sign	Source
MOM12_1	Momentum	$\prod_{j=1}^{11}(1+R_{t-j})-1$	+	Jegadeesh and Titman (1993)
MOM6_1	Momentum	$\prod_{j=1}^{5}(1+R_{t-j})-1$	+	Jegadeesh and Titman (1993)
MOM12_7	Momentum	$\prod_{j=6}^{11}(1+R_{t-j})-1$	+	Novy-Marx (2012)
RESMOM	Momentum	CAPM residuals $t-11..t-1$ / their s.d.; betas from $t-35..t-1$	+	Blitz et al. (2011)
HIGH52	Momentum	$P_t/\max P$ over 252 sessions	+	George and Hwang (2004)
STR	Reversal	R_t	−	Jegadeesh (1990)
LTR	Reversal	$\prod_{j=12}^{35}(1+R_{t-j})-1$	−	De Bondt and Thaler (1985)
SEAS	Seasonality	mean of $R_{t+1-12j}$, $j=1..5$	+	Heston and Sadka (2008)
BETA	Low risk	Dimson beta (m_d, m_{d-1}), 252 sessions	−	Frazzini and Pedersen (2014)
IVOL	Low risk	s.d. of CAPM residuals, month t	−	Ang et al. (2006b)
TVOL	Low risk	s.d. of r_d, month t	−	Ang et al. (2006b)
MAX	Low risk	$\max r_d$, month t	−	Bali et al. (2011)
DBETA	Skewness/downside	beta on days with $m_d < \bar{m}$, 252 sessions	+	Ang et al. (2006a)
SKEW	Skewness/downside	skewness of r_d, 252 sessions	−	Amaya et al. (2015)
ISKEW	Skewness/downside	skewness of CAPM residuals, 252 sessions	−	Boyer et al. (2010)
COSKEW	Skewness/downside	$E[\varepsilon_i\varepsilon_m^2]/(\sqrt{E\varepsilon_i^2}E\varepsilon_m^2)$, 252 sessions	−	Harvey and Siddique (2000)
SIZE	Size/liquidity	log market cap$_t$	−	Banz (1981)
AMIHUD	Size/liquidity	mean $\lvert r_d\rvert/\text{value}_d$, 252 sessions	+	Amihud (2002)
DVOL	Size/liquidity	log mean daily value, 252 sessions	−	Brennan et al. (1998)
TURN	Size/liquidity	$\sum$ value$_{t-11..t}$ / mean cap	−	Datar et al. (1998)
ZERO	Size/liquidity	share of no-trade or zero-return sessions, 252 sessions	+	Lesmond et al. (1999); Bekaert et al. (2007)
SPREAD	Size/liquidity	Corwin–Schultz spread, month t	+	Corwin and Schultz (2012); Amihud and Mendelson (1986)
CVVAL	Size/liquidity	s.d./mean of monthly value, $t-11..t$	−	Chordia et al. (2001)
ABVOL	Size/liquidity	log(daily value in t / daily value $t-12..t-1$)	+	Gervais et al. (2001)
DY	Payout	12-month gross dividends / P_t (payers only)	+	Litzenberger and Ramaswamy (1979)

R_t: monthly total return; r_d: daily total return; m_d: daily MCFTRR return; "252 sessions" are the 252 sessions ending at the last session of month t; "month t" uses the sessions of the formation month. Minimum data: all required months for momentum; at least 20 of 24 months for LTR; at least two years for SEAS; at least 24 months for RESMOM; 120 valid daily returns for the 12-month return-based measures, 120 trading days for HIGH52 and DVOL, and 120 sessions since the first trade for ZERO; 15 valid daily returns for one-month measures; 12 valid two-day pairs for SPREAD. Sign: the direction of predictability reported in the source; long–short portfolios are oriented so that a positive mean confirms it. For ZERO and SPREAD the first source proposes the measure and the second relates illiquidity of that kind to expected returns. DY is sorted among dividend payers only. Appendix Table 12 lists the differences from the source definitions.

4.3 Portfolios

At the end of each month, the shares in the universe with a non-missing characteristic are sorted into quintiles, ties broken by ticker; a factor-month requires at least 25 shares. Portfolios are equally weighted in the baseline, and the long–short return is the top minus the bottom quintile, oriented by the predicted sign. Portfolios are held for one month and rebalanced monthly; returns are gross of costs and taxes. Equal weights are natural here: U100 excludes the illiquid tail, and value weights in a market as concentrated as Russia's would make each quintile a bet on a handful of firms. Capped value weights (Jensen et al., 2023) are a robustness variant.

4.4 Test statistics and the ladder

Let $r_{k,t}$ be the oriented long–short return of characteristic $k = 1, \dots, K$ ($K = 25$) in month t, with mean $\bar{r}_k$ over its n_k available months. The test statistic is $t_k = \bar{r}_k / \widehat{\mathrm{se}}(\bar{r}_k)$ with the Newey and West (1987) variance

$$\widehat{V}(\bar{r}_k) = \frac{1}{n_k^2}\Big[\sum_t e_{k,t}^2 + 2\sum_{j=1}^{L}\Big(1 - \frac{j}{L+1}\Big)\sum_t e_{k,t}e_{k,t-j}\Big], \qquad L = \lfloor 4(T/100)^{2/9} \rfloor = 4, \tag{1}$$

where $e_{k,t} = r_{k,t} - \bar{r}_k$ in available months and zero otherwise. The two-sided p-value is p_k. A *discovery* is a rejection with $t_k > 0$; rejections with the opposite sign are reported separately. The ladder has seven rungs:

R1. Conventional: $t_k > 1.96$.

R2. The Harvey et al. (2016) hurdle: $t_k > 3$.

R3. Holm (1979): with ordered $p_{(1)} \le \cdots \le p_{(K)}$, reject $(1), \dots, (i)$ as long as $p_{(j)} \le \alpha/(K - j + 1)$ for all $j \le i$; this controls the family-wise error rate (FWER).

R4. Benjamini and Hochberg (1995): reject the i^* smallest p-values, $i^* = \max\{i : p_{(i)} \le i\alpha/K\}$; false-discovery rate (FDR) control under independence or positive dependence.

R5. Benjamini and Yekutieli (2001): as R4 with α replaced by $\alpha / \sum_{i=1}^{K} i^{-1}$; FDR control under arbitrary dependence.

R6. Romano and Wolf (2005) stepdown with the studentised statistics. Months are resampled jointly for all factors with the stationary bootstrap of Politis and Romano (1994) (mean block length six months, $B = 10{,}000$), and the centred statistics $t^*_{k,b} = (\bar{r}^*_{k,b} - \bar{r}_k)/\widehat{\mathrm{se}}^*_{k,b}$ are computed with the same Newey–West estimator in every resample. Starting from all hypotheses, the critical value is the 95% quantile of $\max_{k \in A} |t^*_{k,b}|$ over the set A of hypotheses not yet rejected; all $k \in A$ with $|t_k|$ above it are rejected and the step is repeated. Adjusted p-values follow Romano and Wolf (2016). This rung controls the FWER asymptotically while accounting for the dependence and the non-normality of the factor returns; Section 5.2 examines its size at the present sample length.

R7. The Deflated Sharpe Ratio of Bailey and López de Prado (2014). With monthly Sharpe ratio $\widehat{SR}_k$, skewness $\hat{\gamma}_{3,k}$ and kurtosis $\hat{\gamma}_{4,k}$,

$$\mathrm{DSR}_k = \Phi\left(\frac{(\widehat{SR}_k - SR_0)\sqrt{n_k - 1}}{\sqrt{1 - \hat{\gamma}_{3,k}\widehat{SR}_k + \frac{\hat{\gamma}_{4,k}-1}{4}\widehat{SR}_k^2}}\right), \quad SR_0 = \sqrt{\widehat{V}[\widehat{SR}]}\Big[(1-\gamma)\Phi^{-1}\Big(1 - \tfrac{1}{N}\Big) + \gamma\Phi^{-1}\Big(1 - \tfrac{1}{Ne}\Big)\Big], \tag{2}$$

where γ is the Euler–Mascheroni constant, $N = 25$ and $\widehat{V}[\widehat{SR}]$ is the cross-sectional variance of the 25 Sharpe ratios. A factor passes if $\mathrm{DSR}_k > 0.95$.

At the level of the family I report the probability of backtest overfitting (PBO) of choosing the in-sample best of the 25 factors, estimated by combinatorially symmetric cross-validation with 16 blocks (Bailey et al., 2017), and the effective number of independent tests from the eigenvalues λ_i of the factor correlation matrix, both as $1 + (K-1)(1 - \mathrm{Var}(\lambda)/K)$ (Nyholt, 2004) and as $\sum_i [\mathbb{1}(\lambda_i \geq 1) + (\lambda_i - \lfloor \lambda_i \rfloor)]$ (Li and Ji, 2005).

An empirical-Bayes complement. Frequentist corrections ask which individual factors can be declared real while controlling the error rate across the family. Jensen et al. (2023) ask instead what can be inferred about each factor's premium when the premia are themselves drawn from a common distribution. Following them, let $\hat{\mu} = \mu + \varepsilon$, $\varepsilon \sim N(0, \Sigma)$, where Σ is the sample covariance of the factor returns divided by the sample length, and $\mu_k = \mu_0 + c_{j(k)} + w_k$ with cluster effects $c_j \sim N(0, \tau_c^2)$ for the seven clusters of Table 2 (Appendix A discusses the nine labels used in one table of the plan) and idiosyncratic effects $w_k \sim N(0, \tau_w^2)$. The hyperparameters are estimated by maximum marginal likelihood; the posterior of μ is normal, and a factor "replicates" if its 95% posterior interval excludes zero on the positive side.

4.5 Hypotheses

The plan states five hypotheses. H1: each oriented premium is positive ($\mu_k > 0$), evaluated under every rung. H2: the global null that all $\mu_k = 0$, tested with the single-step max-$|t|$ bootstrap. H3: the number of discoveries does not increase along the ladder (descriptive). H4: premia differ between the periods before and after March 2022, for each factor and for the average of the 25—exploratory, given a post-break sample of 55 months. H5: factors that survive the Romano–Wolf rung earn positive returns net of estimated trading costs.

The robustness grid comprises capped value weights, U50, UALL, terciles, a one-session implementation lag (entry at the close of the first trade in month $t+1$), U100 without preferred shares whose ordinary share is also in U100, price-only holding returns, the exclusion of the transition months February and March 2022, CAPM alphas, and point-in-time size and turnover. Every variant is run through the full ladder; the Romano–Wolf test is also repeated with mean block lengths of three and twelve months. For the DSR sensitivity the plan sets $N = 250$, its count of configurations in the grid.

5 Results

5.1 Single-test evidence

Table 3 reports the baseline long–short portfolios and Figure 2 shows their t-statistics against the thresholds of the ladder. The momentum cluster dominates. Nearness to the 52-week high earns 3.08% a month ($t = 6.07$, annualised Sharpe ratio 1.44); six-month momentum 2.19% ($t = 5.33$); residual momentum 1.64% ($t = 4.90$); twelve-month momentum 2.28% ($t = 4.72$); and twelve-to-seven-month momentum 1.54% ($t = 3.24$). Two trading-activity characteristics also clear the conventional bar: abnormal trading volume (1.40%, $t = 3.52$) and the variability of trading value (0.98%, $t = 2.79$). CAPM alphas are close to the raw means because the average market excess return over the sample was small (0.10% a month).

Nothing else clears $t > 1.96$ in the predicted direction. Low-risk portfolios have positive but insignificant means (beta, $t = 1.52$; idiosyncratic volatility, $t = 1.85$; maximum return, $t = 0.13$). Size ($t = 0.41$), Amihud illiquidity ($t = -0.39$), the zero-return proportion ($t = -0.70$), dividend yield ($t = 0.37$), seasonality ($t = -0.41$) and long-term reversal ($t = 0.19$) are statistically zero, and the spread portfolio has the wrong sign ($t = -1.74$). The most striking failure is the one-month reversal: the portfolio that buys last month's losers and sells last month's winners loses 1.96% a month ($t = -4.96$). Section 5.3 returns to this short-term continuation.

Table 3: Baseline long–short portfolios and the ladder of significance

	Long–short return (predicted direction)							Discovery under rung							
Factor	Mean	t	SR	α	$t(\alpha)$	β	Skew	R1	R2	R3	R4	R5	R6	R7	p_{RW}
MOM12_1	2.28	4.72	1.17	2.31	4.98	−0.36	−2.08	✓	✓	✓	✓	✓	✓		0.002
MOM6_1	2.19	5.33	1.34	2.22	5.62	−0.31	−0.43	✓	✓	✓	✓	✓	✓		0.001
MOM12_7	1.54	3.24	0.84	1.57	3.46	−0.35	−1.66	✓	✓	✓	✓	✓			0.058
RESMOM	1.64	4.90	1.28	1.66	5.01	−0.15	−0.43	✓	✓	✓	✓	✓	✓		0.001
HIGH52	3.08	6.07	1.44	3.13	5.95	−0.43	−1.44	✓	✓	✓	✓	✓	✓		0.000
STR	−1.96	−4.96	−1.23	−1.98	−4.76	0.18	0.12	×					×		0.001
LTR	0.08	0.19	0.05	0.07	0.16	0.14	2.13								0.999
SEAS	−0.12	−0.41	−0.10	−0.11	−0.40	−0.09	−0.66								0.999
BETA	0.69	1.52	0.37	0.75	1.74	−0.58	−0.87								0.773
IVOL	0.85	1.85	0.48	0.91	2.18	−0.35	−1.29								0.575
TVOL	0.72	1.46	0.39	0.80	1.87	−0.49	−1.20								0.787
MAX	0.06	0.13	0.04	0.12	0.27	−0.33	−1.53								0.999
DBETA	−0.35	−0.89	−0.22	−0.40	−1.04	0.48	0.80								0.984
SKEW	0.56	1.46	0.38	0.56	1.47	−0.02	−0.54								0.787
ISKEW	0.24	0.59	0.16	0.24	0.62	−0.07	−0.48								0.996
COSKEW	0.06	0.21	0.06	0.05	0.18	0.10	−0.41								0.999
SIZE	0.18	0.41	0.13	0.18	0.40	0.03	0.04								0.999
AMIHUD	−0.16	−0.39	−0.09	−0.14	−0.37	−0.13	1.23								0.999
DVOL	0.24	0.70	0.16	0.26	0.81	−0.19	0.95								0.995
TURN	0.62	1.52	0.46	0.66	1.93	−0.37	−0.44								0.773
ZERO	−0.32	−0.70	−0.20	−0.30	−0.68	−0.16	0.73								0.995
SPREAD	−0.83	−1.74	−0.42	−0.90	−2.01	0.40	1.48								0.644
CVVAL	0.98	2.79	0.63	0.97	2.81	0.04	−1.36	✓			✓				0.137
ABVOL	1.40	3.52	1.02	1.38	3.75	0.20	0.23	✓	✓	✓	✓	✓	✓		0.033
DY	0.16	0.37	0.12	0.17	0.44	−0.21	−0.17								0.999
Number of discoveries (of 25)								7	6	6	7	6	5	0	

Equal-weighted quintile long–short portfolios in U100, oriented by the predicted sign (Table 2). Mean and α in % per month; t-statistics with Newey–West standard errors ($L = 4$). SR: annualised Sharpe ratio. α, β: intercept and slope of the regression on the MCFTRR excess return. Rungs: R1 $t > 1.96$; R2 $t > 3$; R3 Holm; R4 Benjamini–Hochberg; R5 Benjamini–Yekutieli; R6 Romano–Wolf stepdown (stationary bootstrap, mean block 6, $B = 10{,}000$); R7 Deflated Sharpe Ratio > 0.95 with $N = 25$. ✓: discovery in the predicted direction; ×: significant with the opposite sign. p_{RW}: Romano–Wolf adjusted p-value. Holding months 2012:01–2026:09 ($T = 177$). IVOL, TVOL, MAX and SPREAD have 176 months because March 2022 had only six sessions, too few for the one-month daily measures; DY has 176 because fewer than 25 U100 shares had a trailing dividend in May 2014 (Section 3.3).

5.2 Correcting for multiplicity

The global null that no characteristic predicts returns is rejected decisively: the largest $|t|$ is far beyond the bootstrap distribution of the maximum ($p = 0.0002$), confirming H2. The bottom row of Table 3 and the first row of Table 4 show how the discoveries thin out along the ladder: seven at R1, six at R2, six under Holm, seven under Benjamini–Hochberg, six under Benjamini–Yekutieli, five under Romano–Wolf and none under the Deflated Sharpe Ratio. The five Romano–Wolf survivors are the 52-week high,

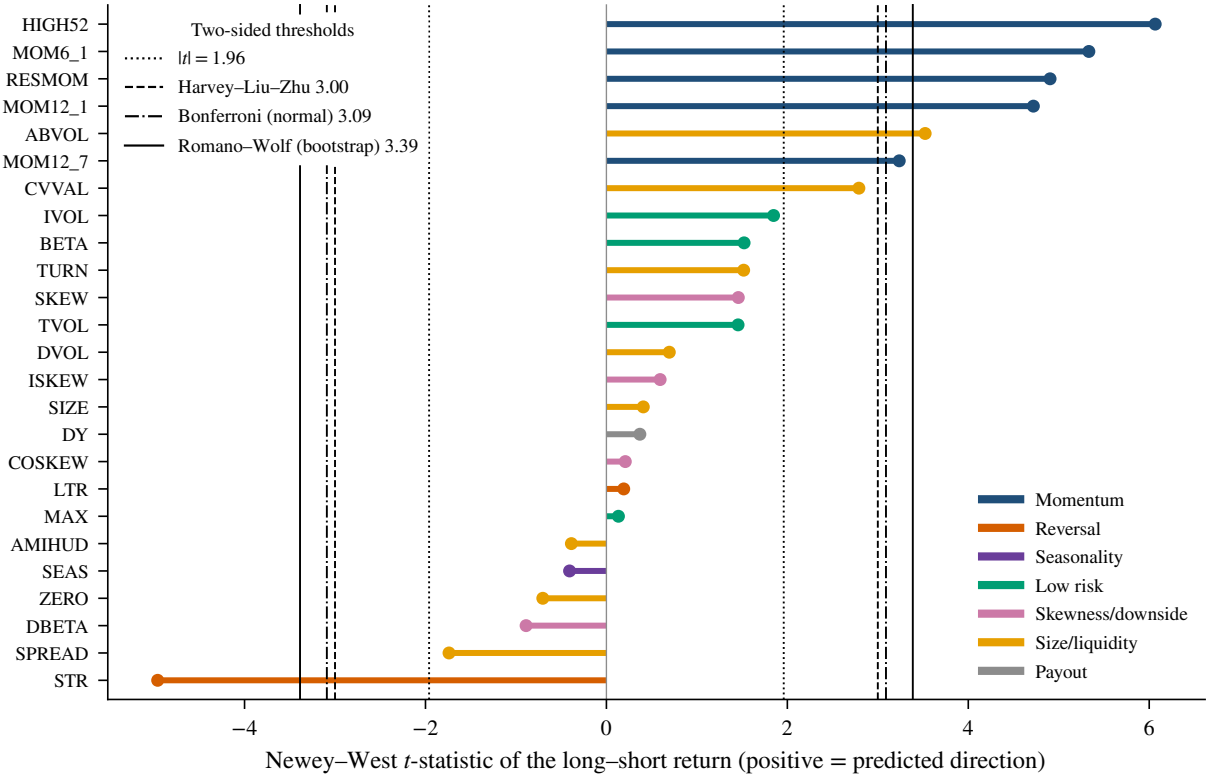


Figure 2: The factor zoo on MOEX. Newey–West t-statistics of the 25 oriented long–short portfolios (baseline), coloured by cluster, against the conventional, Harvey–Liu–Zhu, normal-theory Bonferroni and Romano–Wolf (first-step, bootstrap) thresholds.

six-month momentum, residual momentum, twelve-month momentum and abnormal volume. Twelve-to-seven-month momentum clears the Harvey–Liu–Zhu, Holm and Benjamini–Hochberg bars but not Romano–Wolf ($p_{\mathrm{RW}} = 0.058$); the variability of trading value clears only R1 and Benjamini–Hochberg.

The Romano–Wolf critical value deserves attention. At the first step it equals 3.39, more than the Harvey–Liu–Zhu hurdle and more than 3.09, the Bonferroni bound for 25 two-sided normal tests. Three forces determine it, and simulations separate them (Figure 3). Dependence among the factors lowers the bar: the factors are correlated (the mean absolute pairwise correlation is 0.29, and the first eigenvalue of their correlation matrix is 8.26, or 33% of the total), and standard normal statistics with the factors' correlation matrix have a 95% quantile of the maximum of 2.99 instead of 3.08 for independent ones. The reduction is smaller than the eigenvalue-based effective number of tests suggests (15 by the Li–Ji formula and 22.0 by Nyholt's; 15 independent tests would give 2.93), because moderate correlations matter little in the extreme tail that a family-wise test controls. Estimation noise in the Newey–West standard error raises the bar: with 177 monthly observations the studentised statistics are more dispersed than $N(0,1)$. For i.i.d. normal returns a Newey–West t-statistic with $L = 4$ exceeds 3.09 in absolute value with probability 0.0036 rather than 0.0020, and with the factors' correlation matrix the 95% quantile of the maximum rises to 3.19. The remaining 0.20 reflects features of the actual returns that the bootstrap preserves and the i.i.d. benchmark does not, such as serial dependence and volatility clustering; consistent with this, the critical value rises with the bootstrap block length, to 3.31, 3.39 and 3.55 for mean blocks of three, six and twelve months.

Fat tails of the returns are not responsible: replacing every factor's returns by the normal scores of

their ranks, which sets the kurtosis to three while keeping the dependence, raises the critical value slightly, to 3.47. In the bootstrap the centred statistics have a median standard deviation of 1.07 (between 0.95 and 1.25 across factors) but nearly normal tails (median kurtosis 3.19); rescaling them to unit variance lowers the critical value to 3.11, so most of the excess comes from their dispersion rather than their shape. A longer HAC lag does not help, because it makes the variance estimate noisier: the critical value rises to 3.50 with $L = 8$ and 3.67 with $L = 12$, roughly in step with the i.i.d. benchmark (3.35 and 3.52). Procedures that convert t-statistics into normal p-values—Holm and both FDR rules—therefore understate the multiplicity burden in samples of this length, and the rule of thumb $t > 3$ is not conservative enough. The noise in the variance estimate is generic to any fifteen-year monthly sample, the typical history of an emerging market; the other two components depend on the data. The survivors are the same with mean blocks of three and twelve months except that twelve-to-seven-month momentum enters with the shorter block and abnormal volume drops out with the longer one. Monte Carlo error is modest: with five other seeds the first-step critical value lies between 3.39 and 3.45 and the survivors are the same, with abnormal volume a marginal case (adjusted p between 0.033 and 0.039).

The Romano–Wolf guarantee is asymptotic, and at this sample length the test is mildly liberal. In 4,000 simulated samples under the global null in each of two designs (Appendix B; exploratory), it commits a family-wise error in 5.8% (95% interval 5.1–6.5%) and 7.3% (6.6–8.2%) of the samples, against a nominal 5%. The normal-theory procedures err more often: Holm, which under the global null rejects exactly when Bonferroni does, in 8.0% and 9.8% of the samples, and the $t > 3$ rule in 9.7% and 11.5%. Calibrating the nominal level so that the simulated error rate does not exceed 5% in either design gives $\alpha^* = 0.03$ and a first-step critical value of 3.60. At that level the four momentum survivors and the wrong-signed short-term reversal remain significant, while abnormal volume (adjusted $p = 0.033$) does not (Table 6).

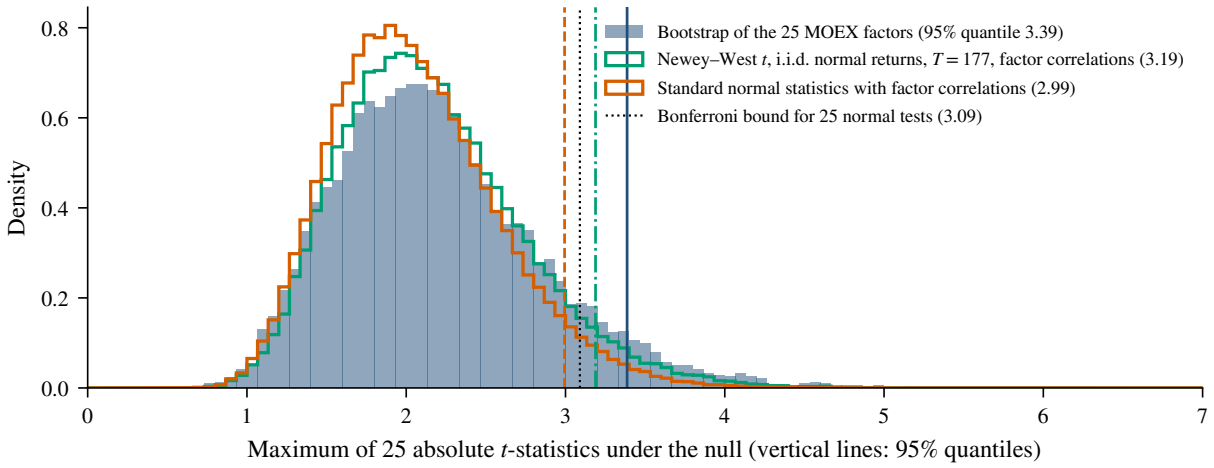


Figure 3: Distribution of the maximum of 25 absolute t-statistics under the null: stationary bootstrap of the 25 MOEX factors (mean block 6, $B = 10{,}000$); Newey–West t-statistics ($L = 4$) computed from i.i.d. normal returns with $T = 177$ and the factors' correlation matrix (100,000 replications); and standard normal statistics with the same correlation matrix. Vertical lines mark the 95% quantiles; the dotted line is the Bonferroni bound for 25 independent normal tests.

No factor passes the pre-specified Deflated Sharpe Ratio. The reason is the benchmark. With $N = 25$ the expected maximum Sharpe ratio under the null is computed from the cross-sectional variance of the 25 estimated Sharpe ratios, and that variance contains the true dispersion between strong momentum

factors and a strongly negative short-term reversal, not only estimation noise. The implied benchmark is an annualised Sharpe ratio of 1.23, so even the best factor, with 1.44, has a DSR of only 0.73. Using the effective number of tests ($N = 15$) lowers the benchmark to 1.09 and changes nothing; using the 250 configurations of the robustness grid raises it to 1.74. As an exploratory diagnostic I replace the benchmark by the expected maximum Sharpe ratio under the dependence-preserving bootstrap null. The benchmark falls to 0.50, and exactly the five Romano–Wolf survivors pass. The closed-form DSR is a useful screen for a researcher's own specification search, where the trials are variations of one strategy; applied to a zoo of genuinely different factors, its variance-based benchmark is extremely conservative.

At the family level, the probability that the in-sample best of the 25 factors underperforms the median factor out of sample is 0.032 (CSCV on the 175 months in which all factors are available). The in-sample winner is a momentum variant in 84% of the 12,870 splits (the 52-week high alone in 47%) and abnormal volume in 14%; out of sample it ranks among the top five of the 25 factors in 78% of splits, and its median rank is fourth. Read literally, H3 fails at one step: Benjamini–Hochberg, which controls the false-discovery rather than the family-wise error rate, admits one factor more than Holm (the variability of trading value). The count never exceeds its conventional level; it falls from seven to six at the Harvey–Liu–Zhu, Holm and Benjamini–Yekutieli rungs, which drop the variability of trading value, and to five at the Romano–Wolf rung, which also drops twelve-to-seven-month momentum.

Table 4: Number of discoveries along the ladder, all specifications

Specification	R1 $t>1.96$	R2 $t>3$	R3 Holm	R4 BH	R5 BY	R6 RW	R7 DSR	Wrong sign (RW)	RW crit. (step 1)
Baseline (U100, EW, quintiles)	7	6	6	7	6	5	0	1	3.39
RW mean block = 3	7	6	6	7	6	6	0	1	3.31
RW mean block = 12	7	6	6	7	6	4	0	1	3.55
Capped value weights	6	4	4	6	4	4	0	0	3.44
U50 (50 most liquid)	6	3	3	6	3	3	0	0	3.31
UALL (all active shares)	10	4	4	8	4	1	0	0	3.64
Terciles	7	6	6	6	6	6	0	1	3.36
One-session implementation lag	8	6	6	6	6	4	0	1	3.32
Preferred duplicates dropped	7	6	6	7	6	6	0	1	3.41
Price-only holding returns	7	4	4	5	4	4	0	1	3.36
Excluding 2022-02, 2022-03	7	7	7	7	7	6	0	1	3.36
Point-in-time SIZE/TURN	7	6	6	7	6	5	0	1	3.42
CAPM alpha (market-hedged)	8	6	6	7	6	6	0	1	3.39
Net of trading costs	5	4	4	4	4	4	0	3	3.39
Pre-2022 (2012-01–2022-02)	7	5	5	6	6	4	0	1	3.47
Post-2022 (2022-03–2026-09)	8	1	1	1	1	0	0	0	3.93

Number of the 25 characteristics declared a discovery in the predicted direction under each rung. Wrong sign (RW): Romano–Wolf rejections with a negative oriented mean. RW crit.: first-step Romano–Wolf critical value. Variants as described in Section 4. CAPM alpha: ladder applied to market-hedged returns. Net of trading costs: returns after the cost model of Section 5.7. Pre/post-2022: ladders estimated separately in each regime (exploratory).

5.3 Short-term continuation

The one-month reversal is one of the oldest anomalies (Jegadeesh, 1990). On MOEX it has the wrong sign: in U100 the winners of month t beat its losers by 1.96% in month $t+1$, a Romano–Wolf-significant rejection of the reversal hypothesis in the opposite direction. The continuation is unlikely to be a mi-

crostructure artefact. Bid–ask bounce and stale prices would produce reversal, not continuation, and the effect is even stronger when the portfolio is entered at the close of the first trade of the holding month ($t = -5.46$; Table 13). It is present before and after 2022 ($t = -4.45$ and -2.04), in U50 ($t = -3.27$) and with value weights ($t = -2.40$), and it survives a Fama–MacBeth control for twelve-month momentum (Section 5.8). It disappears, however, in the broad universe UALL ($t = 0.29$), which adds the least liquid shares.

Medhat and Schmeling (2022) show that the one-month reversal in the United States is confined to low-turnover stocks, while high-turnover stocks display short-term momentum. In an exploratory analysis (Table 5, panel A) I sort UALL into turnover terciles and, within each, form quintiles on the past month's return. Among low-turnover shares last month's winners underperform (-1.23%, $t = -1.90$), among high-turnover shares they outperform (1.04%, $t = 1.58$), and the difference, 2.28% a month, is significant ($t = 2.84$), although neither tercile is significant on its own. Turnover does not explain the continuation in U100, however. Only about half of U100 shares (49%) belong to the top turnover tercile of UALL, and within U100 continuation is strong in both the low-turnover half (1.38%, $t = 3.16$) and the high-turnover half (1.96%, $t = 3.54$), with an insignificant difference ($t = 1.15$; panel B). Continuation is thus a property of the liquid segment as a whole, and weak signs of the usual reversal appear only in the lowest turnover tercile of the broad cross-section. A continuation strategy in U100 turns over 3.10 times its notional a month; net of the cost model of Section 5.7 it would still earn about 0.82% a month ($t = 2.00$) before price impact, which on the illustrative calculation of Section 5.7 would eliminate it at any meaningful size—a borderline and post hoc result.

Table 5: Short-term returns by turnover (exploratory)

	Winners − losers of month t	t
A. All active shares (UALL), turnover terciles		
Low turnover	−1.23	−1.90
Medium turnover	−0.24	−0.38
High turnover	1.04	1.58
High − low	2.28	2.84
B. Liquid universe (U100), turnover halves		
Low turnover	1.38	3.16
High turnover	1.96	3.54
High − low	0.58	1.15

Panel A: within UALL, shares are sorted each month into terciles of turnover (TURN) and, within each tercile, into quintiles of the month-t return. Panel B: the same within U100, with turnover halves. The table reports the mean return in month $t+1$ of the top minus the bottom quintile (winners minus losers, % per month) with Newey–West t-statistics. Positive values indicate continuation. Not part of the pre-analysis plan.

5.4 A Bayesian complement

The empirical-Bayes model gives a much more permissive answer. The estimated prior has mean 0.20% a month, a cluster dispersion $\tau_c = 0.35\%$ and an idiosyncratic dispersion of only $\tau_w = 0.11\%$ (Appendix Table 14, Figure 7). Thirteen of the 25 posterior intervals exclude zero on the positive side, a "replication rate" of 52%. The composition of that set is revealing: it consists of every member of the momentum cluster and every member of the size-and-liquidity cluster, including the Amihud, zero-return and spread portfolios whose own means are negative. With τ_w close to zero the model pools almost completely within

clusters, so the posterior answers whether the average premium of a theme is positive—for liquidity it is, because abnormal volume and the variability of trading value pull the cluster mean up—rather than whether a particular characteristic predicts returns. With the nine cluster labels used in one table of the plan, which separate downside risk from skewness and size from liquidity, the replicated set is the same. Without clusters (exploratory) the model declares 12 factors replicated, but a different set: it includes six portfolios whose own t-statistics are below 1.9 (idiosyncratic and total volatility, skewness, size, trading value and turnover) and excludes twelve-to-seven-month momentum ($t = 3.24$); it ranks the 52-week high below skewness and shrinks the short-term reversal, with its t-statistic of -4.96, to a posterior mean close to zero. These results reflect strong shrinkage across factors whose estimation errors are almost collinear (the correlation is 0.95 between the idiosyncratic- and total-volatility portfolios and 0.87 between twelve-month and twelve-to-seven-month momentum). In both versions the posterior standard deviations (median 0.17% with clusters) are well below the sampling standard errors of the means (median 0.41%), as shrinkage implies, but because the hyperparameters are estimated from only 25 premia and then treated as known, these intervals understate the uncertainty. The Bayesian replication rate is therefore fragile in this setting, and it answers a different question from the frequentist ladder. For the question this paper asks—which specific predictors can be relied on in Russia—the dependence-robust frequentist rung is the more informative.

5.5 Robustness

Table 4 reports the ladder for every specification in the grid, and Appendix Table 13 the t-statistics. The survivors are remarkably stable. Residual momentum survives Romano–Wolf in every full-sample specification of Table 4, including UALL; twelve-month momentum and the 52-week high in all except UALL; six-month momentum in all except value weights, U50 and UALL. Twelve-to-seven-month momentum survives with terciles, without preferred duplicates, for CAPM alphas and with the shortest bootstrap block. Abnormal volume, by contrast, is fragile: it fails in U50, in UALL, with the implementation lag, with price-only returns and with the longest bootstrap block. The failures with the implementation lag and with price-only returns suggest that part of its premium accrues at the very start of the holding month and part through dividends; I do not pursue this further. The variability of trading value survives only when the 2022 transition months are dropped.

In UALL, which adds about 130 less liquid shares, the conventional rung admits ten characteristics—among them size ($t = 2.45$), trading value ($t = 3.11$), the zero-return proportion ($t = 2.47$) and Amihud illiquidity ($t = 2.00$), so the classic small-and-illiquid premia appear once the least liquid shares are included—but the Romano–Wolf critical value rises to 3.64 and only residual momentum survives: the illiquid tail adds noise faster than it adds signal. With value weights and in U50 the momentum core survives, although with fewer variants. Dropping the look-ahead in share counts leaves size and turnover insignificant. Excluding February and March 2022, the most extreme months of the sample, raises rather than lowers the count.

Additional checks (exploratory). Table 6 collects checks added after the first version of the paper, each run through the full ladder. Unobserved delisting losses do not create the momentum premia. Many terminal events are share exchanges or buyouts rather than losses, so a uniform delisting return is a stress

test rather than an estimate. Charging −30%, the value used for performance-related delistings in US data since Shumway (1997), −50%, close to the −55% that Shumway and Warther (1999) estimate for Nasdaq, or −100% once to every terminal event leaves the momentum core and abnormal volume significant. In three of the four momentum portfolios terminal events fall more often in the short leg (in the 52-week-high portfolio, 12 in the long and 20 in the short leg), so assumed losses tend to raise their premia; residual momentum is the exception (9 and 8). The same holds more strongly for the variability of trading value, whose short leg contains 18 of its 21 terminal events; with losses of 100% it also survives. A one-sided bound that charges −100% to every terminal event in a long leg and nothing to those in a short leg, and a two-sided bound that adds a 50% buyout premium to the events in short legs, keep all four momentum survivors, although in the two-sided case residual momentum only just clears the bar ($t = 3.43$ against a critical value of 3.40); abnormal volume falls to $t = 2.95$ and $t = 2.34$. Holding suspended shares until trading resumes changes nothing. In 2013 the price source takes the volumes of some large shares from a thinly traded board; setting their volume-based characteristics to missing for the affected months lowers the t-statistic of abnormal volume to 3.38 and leaves the survivors unchanged (Appendix B). Removing the six depositary-receipt histories (up to six U100 shares a month from 2020) leaves the same five survivors. Re-estimating residual momentum and idiosyncratic volatility with a two-factor model—the market and a value-weighted small-minus-big factor built from all active shares, which inherits the look-ahead in share counts noted in Section 3—lowers the t-statistic of residual momentum from 4.90 to 4.09 and moves that of idiosyncratic volatility from 1.85 to 1.74; the survivors are unchanged. Book equity, and hence a value factor, is not available.

Table 6: Additional checks (exploratory)

	RW	t-statistic						
Specification	crit.	HIGH52	MOM6_1	RESMOM	MOM12_1	ABVOL	RW	Other
Baseline	3.39	6.07	5.33	4.90	4.72	3.52	5	–
Terminal events and suspensions								
Delisting return −30%	3.38	6.14	5.27	4.71	4.75	3.72	5	–
Delisting return −50%	3.37	6.17	5.21	4.56	4.76	3.83	5	–
Delisting return −100%	3.36	6.17	5.02	4.13	4.70	4.03	6	CVVAL
One-sided: −100% long, 0 short	3.39	5.22	4.55	3.71	4.05	2.95	4	–
Two-sided: −100% long, +50% short	3.40	4.71	4.24	3.43	3.68	2.34	4	–
Suspensions held to resumption	3.39	6.07	5.33	4.90	4.72	3.52	5	–
Sample and definitions								
Without receipt histories	3.36	6.04	5.22	4.97	4.70	3.78	5	–
RESMOM, IVOL two-factor	3.42	6.07	5.33	4.09	4.72	3.52	5	–
Size of the test								
Calibrated RW, $\alpha^* = 0.03$	3.60	6.07	5.33	4.90	4.72	3.52	4	–

Newey–West t-statistics of the five baseline survivors and the Romano–Wolf ladder in each specification. RW crit.: first-step Romano–Wolf critical value; RW: number of Romano–Wolf discoveries in the predicted direction; Other: discoveries other than the five survivors. The wrong-signed short-term reversal is a Romano–Wolf rejection in every row. Terminal event: the share stops trading for good during the holding month, or does not trade in it at all and never trades again (predecessors in ISS ticker changes excluded); each event is charged once (Section 3.3). Delisting return: compounded with the return to the last trade, or the whole holding-month return if the share did not trade. One-sided: −100% in long legs, nothing in short legs; two-sided: −100% in long legs and +50% in short legs. Suspensions held: a share that does not trade in the holding month but resumes later earns its return to the first trade after resumption. Without receipts: U100 rebuilt without the six issuers with depositary-receipt histories. Two-factor: RESMOM and IVOL from regressions on the market and a size factor. Calibrated: the nominal level α^* at which the simulated family-wise error rate is at most 5% in both designs of Section 5.2; t-statistics as in the baseline.

5.6 Before and after the 2022 lockout

Table 7 and Figure 4 compare the 122 months before March 2022 with the 55 months after. Four of the five momentum premia are smaller after the break, but all remain large and significant at conventional levels (the 52-week high earns 2.92% a month afterwards, $t = 3.70$). Short-term continuation persists. The clearest change is in the liquidity cluster: trading value, the zero-return proportion and Amihud illiquidity carry negative premia before 2022 and positive premia after (for trading value, -0.27% versus 1.37%, $t(\Delta) = 2.49$), and the low-beta premium rises from 0.44% to 1.23% a month. None of the 25 differences survives the Benjamini–Hochberg correction (the smallest adjusted p-value is 0.16), and no significant change is detected in the average premium of the 25 oriented factors either (0.49% before and 0.70% after, $t(\Delta) = 0.92$). H4 is therefore not supported. Absence of evidence is not evidence of no change, however. The 95% confidence interval for the change in the average premium runs from -0.23 to 0.64 percentage points a month, which includes changes as large as the average premium itself, and a change would have to be about 0.62 percentage points to be detected with 80% probability. For the average of the four momentum survivors, whose premium falls from 2.49% to 1.87% a month, the interval runs from -1.97 to 0.74 points (Table 7 reports the interval for each factor). Within the regimes, the Romano–Wolf test finds four survivors in the pre-break sample (the momentum core) and none in the post-break sample, where with 55 months the critical value rises to 3.93; eight characteristics nonetheless clear $t > 1.96$ after the break.

The comparison is also not a clean experiment. February and March 2022 brought the exit of foreign investors together with sanctions, an emergency increase in the key rate to 20% (Bank of Russia, 2022b), a ban on short sales (Bank of Russia, 2022a), dividend cancellations by several large issuers and a market dominated by individual investors; a before-and-after contrast cannot attribute a change, or its absence, to any one of these. The evidence is consistent with an unchanged average premium, but the post-break sample is too short to rule out economically large changes in either direction.

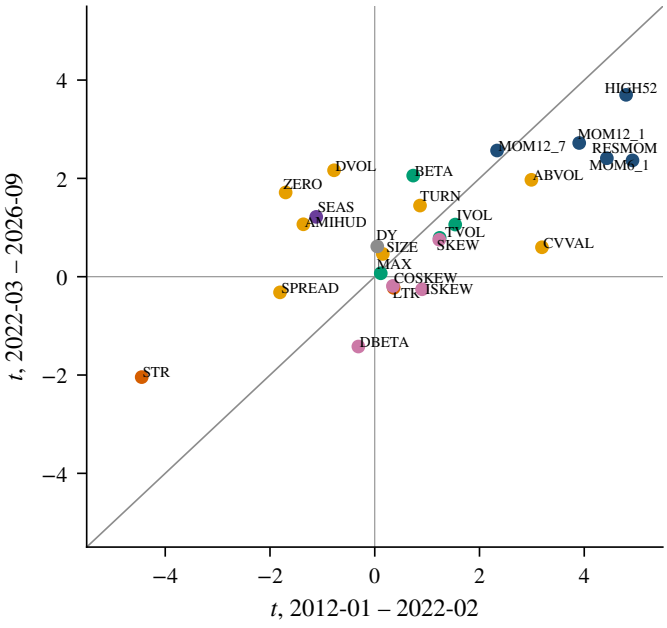


Figure 4: t-statistics of the 25 oriented long–short portfolios before (2012:01–2022:02) and after (2022:03–2026:09) the exclusion of foreign investors, coloured by cluster. The diagonal marks equal t-statistics.

Table 7: Premia before and after the 2022 lockout

	2012-01–2022-02		2022-03–2026-09		Difference			
Factor	Mean	t	Mean	t	Δ	$t(\Delta)$	p_{BH}	95% CI
MOM12_1	2.42	3.90	1.95	2.72	−0.47	−0.51	0.97	[−2.28, 1.33]
MOM6_1	2.55	4.92	1.39	2.37	−1.16	−1.61	0.53	[−2.57, 0.25]
MOM12_7	1.41	2.34	1.82	2.57	0.41	0.44	0.97	[−1.39, 2.20]
RESMOM	1.82	4.43	1.24	2.41	−0.58	−0.92	0.95	[−1.81, 0.65]
HIGH52	3.15	4.80	2.92	3.70	−0.23	−0.23	1.00	[−2.23, 1.76]
STR	−2.12	−4.45	−1.61	−2.04	0.52	0.55	0.97	[−1.32, 2.36]
LTR	0.19	0.36	−0.16	−0.23	−0.35	−0.40	0.97	[−2.05, 1.35]
SEAS	−0.40	−1.12	0.51	1.22	0.90	1.67	0.53	[−0.16, 1.97]
BETA	0.44	0.73	1.23	2.06	0.79	0.97	0.95	[−0.80, 2.39]
IVOL	0.82	1.54	0.94	1.06	0.13	0.12	1.00	[−1.93, 2.18]
TVOL	0.73	1.24	0.72	0.79	−0.01	−0.01	1.00	[−2.14, 2.13]
MAX	0.07	0.12	0.06	0.07	−0.01	−0.01	1.00	[−2.00, 1.98]
DBETA	−0.17	−0.31	−0.77	−1.42	−0.61	−0.82	0.95	[−2.06, 0.84]
SKEW	0.58	1.23	0.51	0.75	−0.07	−0.10	1.00	[−1.57, 1.42]
ISKEW	0.42	0.90	−0.18	−0.26	−0.61	−0.74	0.96	[−2.23, 1.01]
COSKEW	0.13	0.35	−0.09	−0.19	−0.22	−0.36	0.97	[−1.43, 0.99]
SIZE	0.07	0.15	0.42	0.46	0.35	0.33	0.97	[−1.71, 2.40]
AMIHUD	−0.60	−1.36	0.84	1.07	1.44	1.75	0.53	[−0.17, 3.05]
DVOL	−0.27	−0.78	1.37	2.16	1.65	2.49	0.16	[0.35, 2.94]
TURN	0.39	0.86	1.14	1.45	0.74	0.82	0.95	[−1.05, 2.54]
ZERO	−0.94	−1.70	1.07	1.71	2.01	2.49	0.16	[0.43, 3.59]
SPREAD	−1.08	−1.81	−0.27	−0.32	0.81	0.81	0.95	[−1.14, 2.77]
CVVAL	1.23	3.19	0.41	0.60	−0.82	−1.07	0.95	[−2.32, 0.68]
ABVOL	1.43	2.99	1.32	1.97	−0.11	−0.14	1.00	[−1.74, 1.51]
DY	0.02	0.05	0.45	0.62	0.42	0.46	0.97	[−1.36, 2.20]
Average of 25	0.49	3.17	0.70	4.24	0.20	0.92	–	[−0.23, 0.64]

Mean oriented long–short returns (% per month) and Newey–West t-statistics in the two regimes. Δ: coefficient on a post-2022 dummy in a regression of the monthly return on a constant and the dummy (Newey–West, $L = 4$); p_{BH}: Benjamini–Hochberg adjusted p-value across the 25 differences; 95% CI: Δ plus or minus 1.96 Newey–West standard errors. Average of 25: equal-weighted average of the 25 oriented factors.

5.7 Trading costs and implementability

Table 9 charges each trade one half of the Abdi and Ranaldo (2017) close–high–low spread estimate for the formation month plus 5 basis points of fees, and computes turnover from drifted weights. The resulting one-way cost averages 35 to 40 basis points; the estimate is crude, and it ignores price impact. Momentum portfolios turn over about one to one and a half times their notional a month (summing both legs), so the cost drag is 0.4 to 0.6 percentage points a month. I treat H5 as supported when the net mean is significantly positive at the 5% level. All four momentum survivors remain significant net of costs, so H5 holds for them under this cost model: the 52-week high ($t = 5.15$), six-month momentum ($t = 4.00$), twelve-month momentum ($t = 3.97$) and residual momentum ($t = 3.75$). They also survive an exploratory Romano–Wolf test on net returns, and they remain significant at conventional levels when costs are doubled. Their break-even one-way costs are 252, 150, 231 and 148 basis points, respectively. Abnormal volume, which turns over about twice its notional a month, does not survive costs: its net mean, 0.56% a month, is positive but insignificant ($t = 1.40$), so H5 is not supported for it.

Implementability (exploratory). The cost model covers spreads and fees only. Table 8 examines what it leaves out: the availability and cost of short positions, the 2022 short-sale ban and price impact; margin requirements and the financing of net positions are not modelled. Historical data on which shares could be borrowed and at what fee are not available, so the table relies on proxies and break-even values. Between 37 and 52% of each momentum premium comes from the long leg (panel A): measured against the equal-weighted return of all sorted shares, the long leg of the 52-week-high portfolio earns 1.53% a month ($t = 6.65$) and its short leg 1.55% ($t = 4.67$), a split close to the one Israel and Moskowitz (2013) report for US momentum. Restricting the short leg to the 50 most liquid shares, or, from January 2013, the first holding month with an index base in force, to constituents of the MOEX Russia index—rough proxies for the lists of shares that brokers allow clients to sell short—keeps all four momentum premia significant net of spread costs (t between 2.81 and 3.87, and between 2.46 and 4.32). Replacing the short legs by cash in the four months whose short positions would have been opened while short sales were banned, from 24 February to 31 May 2022 (Bank of Russia, 2022a), changes little. A borrow fee on the short leg would have to exceed 15–32% a year to eliminate the net momentum premia; at 10% a year the 52-week high and twelve-month momentum remain significant, six-month momentum is borderline and residual momentum is not. Abnormal volume and short-term continuation, whose net premia are small to begin with, do not survive a borrow fee of 10% a year.

Price impact is the binding constraint. Under the square-root rule of thumb, in which a trade of Q roubles in a share with daily trading value V and daily volatility σ costs about $Y\sigma\sqrt{Q/V}$ with Y of order one (Tóth et al., 2011), the net premia of the equal-weighted momentum portfolios vanish at between 41 and 167 million roubles per leg over the full sample with $Y = 1$, and at between 165 and 667 million with $Y = 0.5$ (panel B). Most of the constraint comes from the early years. In 2012–2015 the median U100 share traded about 7–10 million roubles a day and in 2023–2026 246–294 million; over 2022–2026 the break-even sizes rise to between 101 million and 1.6 billion roubles per leg with $Y = 1$. These calculations are illustrative—the impact coefficient is not estimated for MOEX, and each trade is assumed to be executed within a day—but they show that the gross premia of equal-weighted portfolios overstate what an investor of meaningful size could have earned.

Table 8: Implementability of the survivors (exploratory)

Panel A. Legs and short selling

		Leg vs. all sorted			Break-even	Net,	Net,	Net, shorts only in	
	Gross	Long	Short	Net	fee (% p.a.)	fee 10%	2022 ban	U50	Index‡
HIGH52	3.08	1.53	1.55	2.63	31.6	1.80	2.60	2.34	2.89
	(6.07)	(6.65)	(4.67)	(5.15)		(3.52)	(4.90)	(3.87)	(4.32)
MOM6_1	2.19	0.81	1.37	1.63	19.6	0.80	1.62	1.39	1.29
	(5.33)	(3.60)	(5.64)	(4.00)		(1.96)	(3.78)	(2.81)	(2.46)
RESMOM	1.64	0.86	0.78	1.24	14.9	0.41	1.26	1.23	1.27
	(4.90)	(3.90)	(4.07)	(3.75)		(1.23)	(3.64)	(3.13)	(3.21)
MOM12_1	2.28	1.01	1.26	1.90	22.8	1.07	1.91	1.72	1.93
	(4.72)	(4.70)	(3.99)	(3.97)		(2.23)	(3.86)	(3.38)	(3.67)
ABVOL	1.40	0.53	0.86	0.56	6.7	−0.27	0.61	1.17	0.08
	(3.52)	(2.40)	(4.14)	(1.40)		(−0.68)	(1.56)	(2.09)	(0.14)
STR†	1.96	0.77	1.19	0.82	9.9	−0.01	0.84	0.53	0.54
	(4.96)	(3.07)	(5.70)	(2.00)		(−0.02)	(2.00)	(1.19)	(1.09)

Panel B. Price impact (illustrative)

	Net return				Break-even size (RUB m per leg)			
	RUB 0.1bn per leg		RUB 1bn per leg		$Y = 1$		$Y = 0.5$	
	2012–26	2022–26	2012–26	2022–26	2012–26	2022–26	2012–26	2022–26
HIGH52	0.59	1.88	−3.82	0.51	167	1,581	667	6,323
	(0.93)	(2.18)	(−3.48)	(0.49)				
MOM6_1	−0.91	0.00	−6.40	−1.93	41	101	165	404
	(−1.74)	(0.01)	(−6.10)	(−2.33)				
RESMOM	−0.61	0.29	−4.60	−1.02	45	219	180	875
	(−1.53)	(0.55)	(−5.57)	(−1.60)				
MOM12_1	0.16	1.08	−3.62	−0.11	119	876	475	3,503
	(0.28)	(1.46)	(−4.07)	(−0.14)				
ABVOL	−3.48	−0.65	−12.21	−3.42	2	24	8	97
	(−6.12)	(−0.80)	(−8.27)	(−2.75)				
STR†	−4.81	−1.27	−16.98	−5.38	2	11	9	44
	(−5.37)	(−1.17)	(−7.24)	(−3.06)				

Monthly means in % with Newey–West t-statistics in parentheses, 2012:01–2026:09 unless stated. Panel A: long (short) leg: return of the long leg minus (of all sorted shares minus the short leg), gross of costs, against the equal-weighted return of all shares in the sort. Net: after the spread-based cost model of Section 5.7. Break-even fee: annual borrow fee on the short leg that sets the mean net return to zero. 2022 ban: short legs replaced by cash in the holding months March–June 2022, whose portfolios are formed at the closes of February to May. U50, Index: the short leg holds only those shares of the short quintile that are among the 50 most liquid, or in the MOEX Russia index base in force at the formation date; if no share of the short quintile is eligible, the short leg holds all eligible shares in the sort. ‡Holding months 2013:01–2026:09: the first base takes effect in December 2012 and none is backfilled; over the same months the unrestricted net returns of the four momentum portfolios have t between 3.72 and 5.32. Panel B: net return after spread costs and a square-root impact charge $Y\sigma_i\sqrt{Q_i/V_i}$ per unit traded, where Q_i is the rouble amount traded, V_i the mean daily trading value and σ_i the daily volatility of share i in the formation month, with $Y = 1$ unless stated. When fewer than 15 daily returns are available (as in the formation month March 2022), σ_i is the s.d. of at least five daily returns of the month, or else the share's value for the previous month. Because the charge is proportional to $Y\sqrt{A}$ for a position of A roubles per leg, the break-even size is $A^* = (\bar{n}/\bar{c})^2$, where $\bar{n}$ is the mean net return before impact and $\bar{c}$ the mean impact charge per unit of $\sqrt{A}$. 2022–26: holding months 2022:03–2026:09. STR†: the short-term continuation strategy (winners minus losers of month t).

Table 9: Trading costs

	Gross			Net (1× cost)		Net (2× cost)		Break-even	RW net
Factor	Mean	Turnover	Cost	Mean	t	Mean	t	(bp, one-way)	
MOM12_1	2.28	0.99	0.37	1.90	3.97	1.53	3.20	231	✓
MOM6_1	2.19	1.46	0.56	1.63	4.00	1.07	2.64	150	✓
MOM12_7	1.54	1.35	0.51	1.02	2.15	0.51	1.07	114	
RESMOM	1.64	1.11	0.40	1.24	3.75	0.84	2.56	148	✓
HIGH52	3.08	1.23	0.45	2.63	5.15	2.19	4.22	252	✓
STR	−1.96	3.10	1.14	−3.10	−7.91	−4.24	−10.55	–	
LTR	0.08	0.78	0.29	−0.20	−0.49	−0.49	−1.16	10	
SEAS	−0.12	3.07	1.14	−1.26	−4.16	−2.40	−7.29	–	
BETA	0.69	0.71	0.31	0.38	0.83	0.07	0.15	97	
IVOL	0.85	2.32	0.87	−0.02	−0.04	−0.89	−1.85	37	
TVOL	0.72	2.32	0.90	−0.18	−0.36	−1.08	−2.13	31	
MAX	0.06	2.69	1.04	−0.97	−2.00	−2.01	−4.05	2	
DBETA	−0.35	0.86	0.37	−0.72	−1.81	−1.09	−2.69	–	
SKEW	0.56	0.86	0.36	0.20	0.51	−0.16	−0.41	65	
ISKEW	0.24	0.85	0.36	−0.12	−0.30	−0.48	−1.14	28	
COSKEW	0.06	0.98	0.39	−0.33	−1.01	−0.73	−2.03	7	
SIZE	0.18	0.38	0.13	0.05	0.11	−0.09	−0.20	48	
AMIHUD	−0.16	0.45	0.19	−0.35	−0.86	−0.54	−1.33	–	
DVOL	0.24	0.46	0.19	0.05	0.16	−0.13	−0.38	51	
TURN	0.62	0.40	0.14	0.48	1.17	0.34	0.83	155	
ZERO	−0.32	0.68	0.26	−0.58	−1.28	−0.84	−1.85	–	
SPREAD	−0.83	2.25	0.84	−1.67	−3.50	−2.51	−5.18	–	
CVVAL	0.98	0.79	0.30	0.68	1.90	0.38	1.04	124	
ABVOL	1.40	2.08	0.84	0.56	1.40	−0.27	−0.66	67	
DY	0.16	0.80	0.26	−0.10	−0.24	−0.36	−0.85	20	

Gross mean, net means and costs in % per month. Turnover: average monthly sum over both legs of $|w_{\text{new}} - w_{\text{drifted}}|$. One-way cost per share-month: half of the Abdi and Ranaldo (2017) spread estimate for the formation month plus 5 bp (cross-sectional median if unavailable). Break-even: one-way cost in basis points that sets the mean net return to zero; "–" for negative gross means. RW net: discovery under the Romano–Wolf rung for net returns (exploratory). Newey–West t-statistics.

5.8 Fama–MacBeth regressions

Table 10 reports monthly cross-sectional regressions of next-month returns on cross-sectional ranks scaled to $[-0.5, 0.5]$ and oriented, so that a slope is the return spread between the highest- and lowest-ranked share (Fama and MacBeth, 1973). Univariate slopes reproduce the portfolio sorts (the 52-week high has $t = 6.23$). When all 25 ranks enter jointly, as the plan prescribes, no slope is significant (the largest $|t|$ is 1.83): the five momentum signals are collinear and share their explanatory power. Two exploratory specifications illustrate this. With one canonical characteristic per cluster, twelve-month momentum ($t = 5.22$) and short-term continuation ($t = -3.46$) are the only significant predictors. With the Romano–Wolf survivors and short-term continuation together, twelve-month momentum keeps $t = 2.09$, the 52-week high $t = 1.76$ and continuation $t = -2.37$, while abnormal volume falls to $t = 1.02$. The momentum survivors are therefore best read as heavily overlapping measurements of trend; abnormal volume partly overlaps with short-term continuation.

The time series of the portfolio returns agree (exploratory). The four momentum survivors have pairwise correlations between 0.40 and 0.78, and the first principal component of their correlation matrix accounts for 73% of the total. Regressed on the other three, six-month momentum ($t(\alpha) = 0.80$) and twelve-month momentum ($t(\alpha) = -0.92$) are spanned, while the 52-week high ($t(\alpha) = 2.33$) and residual momentum ($t(\alpha) = 2.27$) keep alphas that are significant only at the conventional level. An equal-weighted composite of the four earns 2.30% a month ($t = 6.10$). Regressed on the 52-week high and residual momentum alone, six- and twelve-month momentum have $t(\alpha) = 0.47$ and -0.81. Abnormal volume is nearly uncorrelated with the momentum survivors (correlations of 0.19 or less), and its alpha against them has $t = 1.93$. These tests do not establish the exact number of distinct effects, since an insignificant incremental alpha is also what weak power would produce, but they show that the four survivors are largely redundant: the robust evidence is one momentum family, plus a fragile high-volume premium.

5.9 Statistical power

How much could fifteen years of Russian data reveal? I simulate samples of 177 months by resampling the demeaned 25-factor return matrix with the stationary bootstrap, which preserves the dependence and the tails, and inject a true premium into one factor at a time. Table 11 and Figure 5 report the probability that the injected factor is discovered, averaged over the 25 factors. A factor with an annual Sharpe ratio of 0.5 clears $t > 1.96$ about half the time but survives Romano–Wolf with probability of only 0.13; at 0.75 the Romano–Wolf power is 0.39; at 1.0, 0.72; and only at 1.25 does it exceed 0.9. The survivors have Sharpe ratios of 1.02 to 1.44, within the detectable range, although abnormal volume lies near its lower edge. Many anomalies documented in the United States have long–short Sharpe ratios well below one, and the non-surviving low-risk, size, liquidity and skewness portfolios here have Sharpe ratios between roughly -0.4 and 0.6. Even the conventional test detects a premium with a Sharpe ratio of 0.75 with probability 0.81. Premia that large are therefore unlikely for the factors whose estimated Sharpe ratios are near or below zero, but not for the low-risk, skewness and turnover portfolios, whose t-statistics lie between 1.4 and 1.9; and premia with Sharpe ratios of 0.5 or less cannot be ruled out for any non-replicated factor. Their failure to replicate on MOEX is weak evidence against their existence.

Table 10: Fama–MacBeth regressions

	Univariate		All 25 jointly		One per cluster*		Survivors + STR*	
Characteristic	Slope	t	Slope	t	Slope	t	Slope	t
MOM12_1	2.63	4.68	1.75	0.98	2.63	5.22	1.67	2.09
MOM6_1	2.23	4.60	0.49	0.42			0.06	0.10
MOM12_7	1.89	3.68	0.09	0.06				
RESMOM	1.96	4.66	0.29	0.28			0.30	0.52
HIGH52	3.45	6.23	0.98	1.07			1.32	1.76
STR	−2.31	−4.75	−0.87	−1.54	−1.85	−3.46	−1.21	−2.37
LTR	0.05	0.10	0.63	0.80				
SEAS	0.09	0.26	−0.46	−0.98	−0.30	−0.87		
BETA	0.89	1.80	1.28	1.53	0.43	0.56		
IVOL	0.86	1.58	1.72	1.57				
TVOL	0.93	1.61	−0.05	−0.04				
MAX	0.29	0.52	−1.00	−1.05				
DBETA	−0.37	−0.82	0.47	0.61	0.21	0.34		
SKEW	0.50	1.12	0.31	0.29				
ISKEW	0.34	0.71	0.33	0.32				
COSKEW	0.20	0.56	0.51	0.91				
SIZE	0.05	0.09	1.51	1.40	0.54	1.14		
AMIHUD	−0.33	−0.70	−0.95	−0.50				
DVOL	0.10	0.24	0.10	0.06				
TURN	0.72	1.55	1.26	1.83				
ZERO	−0.34	−0.72	0.03	0.06				
SPREAD	−0.85	−1.59	−0.27	−0.68				
CVVAL	1.00	2.52	0.48	0.83				
ABVOL	1.52	3.49	0.42	0.98			0.46	1.02
DY	0.20	0.42	0.61	1.39	0.04	0.09		

Monthly cross-sectional OLS regressions in U100 of the month-$t+1$ return on cross-sectional ranks of the characteristics scaled to $[-0.5, 0.5]$ and oriented by the predicted sign; missing ranks are set to zero in the multivariate regressions. Time-series averages of the slopes in % per month with Newey–West t-statistics ($L = 4$). Univariate and "All 25 jointly" are pre-specified; columns marked * are exploratory.

Table 11: Power of the ladder

True annual SR	R1 t>1.96	R2 t>3	R3 Holm	R4 BH	R5 BY	R6 RW
0.25	0.18	0.04	0.03	0.04	0.02	0.03
0.50	0.50	0.18	0.16	0.17	0.10	0.13
0.75	0.81	0.48	0.45	0.45	0.33	0.39
1.00	0.96	0.79	0.76	0.77	0.65	0.72
1.25	0.99	0.94	0.93	0.94	0.89	0.91
1.50	1.00	0.99	0.99	0.99	0.97	0.98

Probability that a single factor with the given true annualised Sharpe ratio is discovered (correct sign) when the other 24 factors have zero premia. Samples of 177 months are drawn with the stationary bootstrap (mean block 6) from the demeaned baseline factor returns; 300 outer resamples, each with 499 inner resamples for Romano–Wolf; averages over the 25 factors.

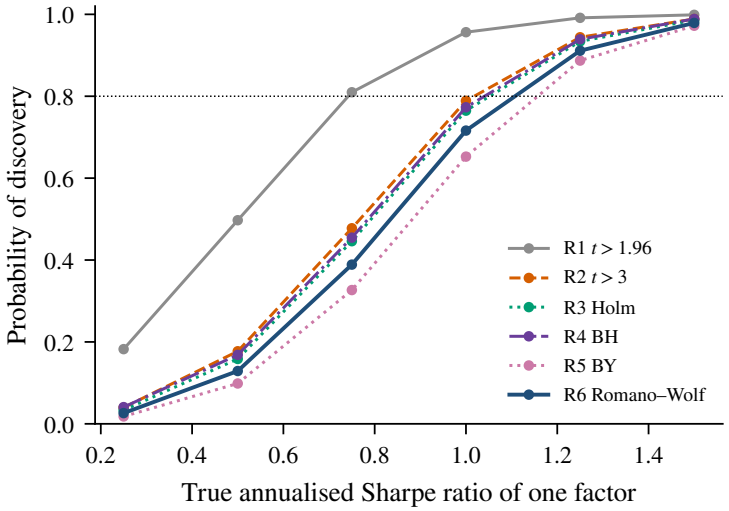


Figure 5: Probability of discovering one true factor as a function of its annualised Sharpe ratio, by rung.

6 Discussion

What survives on MOEX. The evidence supports one robust anomaly family: intermediate-horizon price momentum, measured by past six- and twelve-month returns, residual returns or nearness to the 52-week high. These are largely redundant measurements of one family, not four discoveries. The premia are large (Sharpe ratios above one), survive the most demanding correction in nearly every full-sample variation of the protocol (residual momentum in all of them), a size-calibrated version of that correction, bounds for unobserved delisting losses, spread-based trading costs and restrictions of the short leg to liquid or index shares, and remain positive and significant at conventional levels after 2022, although in the 55-month post-break sample no factor clears the Romano–Wolf bar. Their capacity in equal-weighted form is small, so the premia are better read as evidence about price formation than as a measure of what a large investor could earn. Short-term continuation among liquid shares is a second, less robust regularity: it passes the Romano–Wolf test in eight of the eleven full-sample specifications (not with value weights, in U50 or in UALL), and its net-of-cost return is borderline. Their cumulative returns (Figure 6) rise steadily except for a sharp momentum crash in February 2015, when the losers of the 2014 rouble crisis rebounded and twelve-month momentum lost 44.9% in a month, an episode of the kind described by Daniel and Moskowitz (2016). The finding contrasts with the absence of momentum in Eastern Europe over 1990–2011 reported by Cakici et al. (2013) and with the contrarian profits at longer horizons in Teplova and Mikova (2014), and it agrees with the significant momentum premium of Ovechkin and Boldyreva (2022) and the after-cost momentum profits of Nazarova and Leshchev (2023). A market dominated by individual investors, with short sales restricted at times of stress and episodic sanctions shocks, is a natural place for slow information diffusion and under-reaction to persist.

What does not survive. The implemented low-risk, size, liquidity, skewness, seasonality, long-term reversal and dividend-yield signals do not survive in the liquid universe. For liquidity this matches Teplova and Gurov (2022) and Gurov (2023), who find little cross-sectional evidence of an illiquidity premium on MOEX; for size it matches Ovechkin and Boldyreva (2022). Three qualifications limit what this means. The conclusion concerns these implementations, which differ from several source definitions (Appendix

Table 12); it concerns the 100 most liquid shares, and the classic size and illiquidity premia do appear at the conventional level once the least liquid shares are included (UALL); and it concerns a fifteen-year sample in which premia with Sharpe ratios of 0.5 or less would be detected at most about half the time by the conventional test and rarely by the robust one. A failure to survive is therefore weak evidence that an anomaly is absent from the Russian market.

Implications for empirical work on Russia and other emerging markets. First, in samples of fifteen years or less the familiar $t > 3$ hurdle is not conservative enough: autocorrelation-robust t-statistics are more dispersed than the normal distribution, and the bootstrap Romano–Wolf critical value ranges from 3.3 to 3.6 across specifications and approaches 4 in a 55-month subsample. Researchers working with short emerging-market histories should use bootstrap critical values for the studentised statistics rather than normal p-values, and should check the size of the bootstrap test by simulation, because even it is somewhat liberal at these sample lengths. Second, the closed-form Deflated Sharpe Ratio should not be applied to a zoo of heterogeneous factors with the cross-factor variance of Sharpe ratios as its benchmark; a bootstrap null benchmark gives sensible answers. Third, survivorship matters: about a third of the shares that were ever among the 100 most liquid are no longer listed, and even a source that includes delisted shares can miss delisted instruments of another type, as the depositary receipts missing here show. Fourth, Bayesian and frequentist replication rates answer different questions; reporting both, as here, makes the difference explicit.

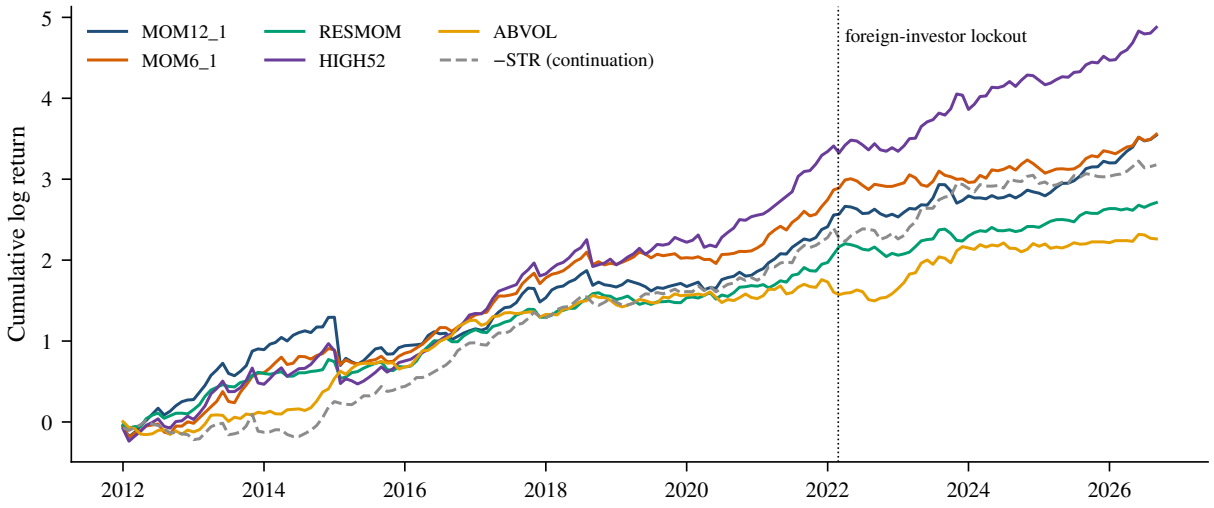


Figure 6: Cumulative log returns of the five Romano–Wolf survivors and of short-term continuation (the negative of the reversal portfolio), baseline specification, 2012:01–2026:09. The dotted line marks the February 2022 suspension.

Limitations. The study covers only characteristics that can be built from market data and dividends; the value, profitability, investment and accrual families, which carry much of the US evidence, require accounting data that are not available in open sources for the full cross-section including delisted firms. Market capitalisation is point-in-time only for index constituents. Dividend coverage is thinner in the early years—the number of dividend-paying securities recorded rises from 76 in 2010 to at least 100 in every year from 2011—which affects only the formation windows of the longest-horizon characteristics in the first months; the momentum core survives with price-only returns, so it does not hinge on dividends.

Depositary receipts are covered only in part (Section 3.2). The final payoffs of delisted shares are not observed and are bounded rather than measured. Historical data on which shares could be sold short and at what fee are not available, so the implementability analysis relies on proxies; it does not model margin requirements or the financing of net positions, and its price-impact calculation uses a rule of thumb rather than estimates for MOEX. And the post-2022 sample is short.

7 Conclusion

A pre-specified test of 25 anomalies on fifteen years of Moscow Exchange share data free of survivorship bias finds that intermediate-horizon price momentum is the only anomaly family that survives a dependence-robust bootstrap correction for multiple testing, nearly every variation of the protocol, bounds for unobserved delisting losses and spread-based trading costs; its four surviving measures are largely redundant, and it remains significant at conventional levels after the 2022 exclusion of foreign investors. Liquid Russian shares also exhibit short-term continuation rather than reversal. The implemented low-risk, size, liquidity, skewness, seasonality and payout signals do not survive, but the sample cannot reliably detect premia with Sharpe ratios much below one, so this is weak evidence that those anomalies are absent. Methodologically, the Russian evidence shows that with fifteen years of monthly data the bootstrap stepdown bar is higher than the familiar $t > 3$ rule, because autocorrelation-robust t-statistics are more dispersed than the normal distribution; corrections based on normal p-values therefore overstate significance in samples of this length, and even the bootstrap test is somewhat liberal. Extending the zoo to accounting-based characteristics, measuring short-sale costs and price impact on MOEX directly, and revisiting the post-2022 regime as its sample lengthens are natural next steps.

Data availability

All data are public. The replication package is available at `https://github.com/Sergpreneur/moex-factor-zoo` (release v1.0). It contains the pre-analysis plan with its hash, timestamp and version history, the list of deviations, the code that rebuilds every number, table and figure from the raw data, the monthly factor return series (also as CSV files), the tests, and a provenance record of the data sources with their commit hashes; the raw data are downloaded from their public sources by a script.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used Claude (Anthropic) to assist with code development, data processing and drafting of the manuscript. The author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

Acknowledgements

I thank the maintainers of the public MOEX data mirrors used in this study, in particular S. Andreev for the `moex-momentum` collection released under the MIT licence.

References

Abdi, F. and Ranaldo, A. (2017). A simple estimation of bid-ask spreads from daily close, high, and low prices. *Review of Financial Studies*, 30(12):4437–4480.

Abramov, A., Chernova, M., and Akhmetzyanov, T. (2025). Evolution of factor pricing models and their application in the Russian financial market. *Journal of Corporate Finance Research*, 19(2):67–81.

Abramov, A. E., Radygin, A. D., and Chernova, M. I. (2019). Pricing models of shares of Russian companies and their practical application. *Voprosy Ekonomiki*, (3):48–76. In Russian.

Amaya, D., Christoffersen, P., Jacobs, K., and Vasquez, A. (2015). Does realized skewness predict the cross-section of equity returns? *Journal of Financial Economics*, 118(1):135–167.

Amihud, Y. (2002). Illiquidity and stock returns: cross-section and time-series effects. *Journal of Financial Markets*, 5(1):31–56.

Amihud, Y. and Mendelson, H. (1986). Asset pricing and the bid-ask spread. *Journal of Financial Economics*, 17(2):223–249.

Anatolyev, S. (2008). A 10-year retrospective on the determinants of Russian stock returns. *Research in International Business and Finance*, 22(1):56–67.

Andreev, S. (2026). moex-momentum: survivorship-free MOEX ISS share prices, dividends and splits. GitHub repository, commit 240a69d, MIT licence.

Ang, A., Chen, J., and Xing, Y. (2006a). Downside risk. *Review of Financial Studies*, 19(4):1191–1239.

Ang, A., Hodrick, R. J., Xing, Y., and Zhang, X. (2006b). The cross-section of volatility and expected returns. *Journal of Finance*, 61(1):259–299.

Bailey, D. H., Borwein, J. M., López de Prado, M., and Zhu, Q. J. (2017). The probability of backtest overfitting. *Journal of Computational Finance*, 20(4):39–69.

Bailey, D. H. and López de Prado, M. (2014). The deflated Sharpe ratio: Correcting for selection bias, backtest overfitting, and non-normality. *Journal of Portfolio Management*, 40(5):94–107.

Bali, T. G., Cakici, N., and Whitelaw, R. F. (2011). Maxing out: Stocks as lotteries and the cross-section of expected returns. *Journal of Financial Economics*, 99(2):427–446.

Bank of Russia (2022a). The Bank of Russia limits investor risks in margin trading. Press release, 20 May 2022. In Russian.

Bank of Russia (2022b). Statement by Bank of Russia governor Elvira Nabiullina in follow-up to Board of Directors meeting on 28 February 2022. 28 February 2022.

Bank of Russia (2022c). Trading on Moscow Exchange on 24 March. Press release, 23 March 2022.

Banz, R. W. (1981). The relationship between return and market value of common stocks. *Journal of Financial Economics*, 9(1):3–18.

Bekaert, G., Harvey, C. R., and Lundblad, C. (2007). Liquidity and expected returns: Lessons from emerging markets. *Review of Financial Studies*, 20(6):1783–1831.

Benjamini, Y. and Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society. Series B (Methodological)*, 57(1):289–300.

Benjamini, Y. and Yekutieli, D. (2001). The control of the false discovery rate in multiple testing under dependency. *Annals of Statistics*, 29(4):1165–1188.

Blitz, D., Huij, J., and Martens, M. (2011). Residual momentum. *Journal of Empirical Finance*, 18(3):506–521.

Boyer, B., Mitton, T., and Vorkink, K. (2010). Expected idiosyncratic skewness. *Review of Financial Studies*, 23(1):169–202.

Brennan, M. J., Chordia, T., and Subrahmanyam, A. (1998). Alternative factor specifications, security characteristics, and the cross-section of expected stock returns. *Journal of Financial Economics*, 49(3):345–373.

Cakici, N., Fabozzi, F. J., and Tan, S. (2013). Size, value, and momentum in emerging market stock returns. *Emerging Markets Review*, 16:46–65.

Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 11(2):207–264.

Chordia, T., Goyal, A., and Saretto, A. (2020). Anomalies and false rejections. *Review of Financial Studies*, 33(5):2134–2179.

Chordia, T., Subrahmanyam, A., and Anshuman, V. R. (2001). Trading activity and expected stock returns. *Journal of Financial Economics*, 59(1):3–32.

Cochrane, J. H. (2011). Presidential address: Discount rates. *Journal of Finance*, 66(4):1047–1108.

Corwin, S. A. and Schultz, P. (2012). A simple way to estimate bid-ask spreads from daily high and low prices. *Journal of Finance*, 67(2):719–760.

Daniel, K. and Moskowitz, T. J. (2016). Momentum crashes. *Journal of Financial Economics*, 122(2):221–247.

Datar, V. T., Naik, N. Y., and Radcliffe, R. (1998). Liquidity and stock returns: An alternative test. *Journal of Financial Markets*, 1(2):203–219.

De Bondt, W. F. M. and Thaler, R. (1985). Does the stock market overreact? *Journal of Finance*, 40(3):793–805.

Dimson, E. (1979). Risk measurement when shares are subject to infrequent trading. *Journal of Financial Economics*, 7(2):197–226.

Fama, E. F. and MacBeth, J. D. (1973). Risk, return, and equilibrium: Empirical tests. *Journal of Political Economy*, 81(3):607–636.

Feng, G., Giglio, S., and Xiu, D. (2020). Taming the factor zoo: A test of new factors. *Journal of Finance*, 75(3):1327–1370.

Frazzini, A. and Pedersen, L. H. (2014). Betting against beta. *Journal of Financial Economics*, 111(1):1–25.

George, T. J. and Hwang, C.-Y. (2004). The 52-week high and momentum investing. *Journal of Finance*, 59(5):2145–2176.

Gervais, S., Kaniel, R., and Mingelgrin, D. H. (2001). The high-volume return premium. *Journal of Finance*, 56(3):877–919.

Giglio, S., Liao, Y., and Xiu, D. (2021). Thousands of alpha tests. *Review of Financial Studies*, 34(7):3456–3496.

Goriaev, A. and Zabotkin, A. (2006). Risks of investing in the Russian stock market: Lessons of the first decade. *Emerging Markets Review*, 7(4):380–397.

Griffin, J. M., Ji, X., and Martin, J. S. (2003). Momentum investing and business cycle risk: Evidence from pole to pole. *Journal of Finance*, 58(6):2515–2547.

Gurov, S. (2023). Illiquidity effects in the Russian stock market. *HSE Economic Journal*, 27(1):78–102. In Russian.

Hanauer, M. X. and Lauterbach, J. G. (2019). The cross-section of emerging market stock returns. *Emerging Markets Review*, 38:265–286.

Hansen, P. R. (2005). A test for superior predictive ability. *Journal of Business & Economic Statistics*, 23(4):365–380.

Harvey, C. R. and Liu, Y. (2020). False (and missed) discoveries in financial economics. *Journal of Finance*, 75(5):2503–2553.

Harvey, C. R., Liu, Y., and Zhu, H. (2016). … and the cross-section of expected returns. *Review of Financial Studies*, 29(1):5–68.

Harvey, C. R. and Siddique, A. (2000). Conditional skewness in asset pricing tests. *Journal of Finance*, 55(3):1263–1295.

Heston, S. L. and Sadka, R. (2008). Seasonality in the cross-section of stock returns. *Journal of Financial Economics*, 87(2):418–445.

Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70.

Hou, K., Xue, C., and Zhang, L. (2020). Replicating anomalies. *Review of Financial Studies*, 33(5):2019–2133.

Israel, R. and Moskowitz, T. J. (2013). The role of shorting, firm size, and time on market anomalies. *Journal of Financial Economics*, 108(2):275–301.

Jacobs, H. and Müller, S. (2020). Anomalies across the globe: Once public, no longer existent? *Journal of Financial Economics*, 135(1):213–230.

Jegadeesh, N. (1990). Evidence of predictable behavior of security returns. *Journal of Finance*, 45(3):881–898.

Jegadeesh, N. and Titman, S. (1993). Returns to buying winners and selling losers: Implications for stock market efficiency. *Journal of Finance*, 48(1):65–91.

Jensen, T. I., Kelly, B., and Pedersen, L. H. (2023). Is there a replication crisis in finance? *Journal of Finance*, 78(5):2465–2518.

Kinnunen, J. and Martikainen, M. (2017). Expected returns and idiosyncratic risk: Industry-level evidence from Russia. *Emerging Markets Finance and Trade*, 53(11):2528–2544.

Lesmond, D. A., Ogden, J. P., and Trzcinka, C. A. (1999). A new estimate of transaction costs. *Review of Financial Studies*, 12(5):1113–1141.

Li, J. and Ji, L. (2005). Adjusting multiple testing in multilocus analyses using the eigenvalues of a correlation matrix. *Heredity*, 95(3):221–227.

Litzenberger, R. H. and Ramaswamy, K. (1979). The effect of personal taxes and dividends on capital asset prices: Theory and empirical evidence. *Journal of Financial Economics*, 7(2):163–195.

Lo, A. W. and MacKinlay, A. C. (1990). Data-snooping biases in tests of financial asset pricing models. *Review of Financial Studies*, 3(3):431–467.

McLean, R. D. and Pontiff, J. (2016). Does academic research destroy stock return predictability? *Journal of Finance*, 71(1):5–32.

Medhat, M. and Schmeling, M. (2022). Short-term momentum. *Review of Financial Studies*, 35(3):1480–1526.

Moscow Exchange (2022). Press release on trading in all Russian shares from 28 March 2022. Press release, March 2022.

Moscow Exchange (2023). Transition to the T+1 settlement cycle for shares and bonds. Press release, July 2023.

Moscow Exchange (2024). Trading results for 2023. Press release, 12 January 2024.

Nazarova, V. V. and Leshchev, S. I. (2023). Study of the momentum effect in the price dynamics of highly liquid shares on the Russian securities market. *Financial Journal*, 15(1):58–73. In Russian.

Newey, W. K. and West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3):703–708.

Novy-Marx, R. (2012). Is momentum really momentum? *Journal of Financial Economics*, 103(3):429–453.

Nyholt, D. R. (2004). A simple correction for multiple testing for single-nucleotide polymorphisms in linkage disequilibrium with each other. *American Journal of Human Genetics*, 74(4):765–769.

Ovechkin, D. V. and Boldyreva, N. B. (2022). Econometric assessment of risk premiums in the Russian stock market. *Tyumen State University Herald. Social, Economic, and Law Research*, 8(1):331–347.

Politis, D. N. and Romano, J. P. (1994). The stationary bootstrap. *Journal of the American Statistical Association*, 89(428):1303–1313.

President of the Russian Federation (2022a). Decree no. 81 of 1 March 2022 "on additional temporary economic measures to ensure the financial stability of the Russian Federation". In Russian.

President of the Russian Federation (2022b). Decree no. 95 of 5 March 2022 "on the temporary procedure for fulfilling obligations to certain foreign creditors". In Russian.

quantviews (2026). moexutils: MOEX ISS utilities and index calculation bases. GitHub repository, commit 9742060.

Romano, J. P. and Wolf, M. (2005). Stepwise multiple testing as formalized data snooping. *Econometrica*, 73(4):1237–1282.

Romano, J. P. and Wolf, M. (2016). Efficient computation of adjusted p-values for resampling-based stepdown multiple testing. *Statistics & Probability Letters*, 113:38–40.

Rouwenhorst, K. G. (1999). Local return factors and turnover in emerging stock markets. *Journal of Finance*, 54(4):1439–1464.

Russian Federation (2013). Federal law no. 379-FZ of 21 December 2013 "on amendments to certain legislative acts of the Russian Federation". Art. 3 amends Art. 42(5) of the Federal Law "On Joint-Stock Companies" with effect from 1 January 2014. In Russian.

Shumway, T. (1997). The delisting bias in CRSP data. *Journal of Finance*, 52(1):327–340.

Shumway, T. and Warther, V. A. (1999). The delisting bias in CRSP's Nasdaq data and its implications for the size effect. *Journal of Finance*, 54(6):2361–2379.

Teplova, T. and Gurov, S. (2022). New evidence on the impact of implicit trading costs on asset prices in the Russian stock market. *Applied Economics*, 54(51):5943–5955.

Teplova, T. and Shutova, E. (2011). A higher moment downside framework for conditional and unconditional CAPM in the Russian stock market. *Eurasian Economic Review*, 1(2):157–178.

Teplova, T. V. and Mikova, E. S. (2014). Company size, trading activity and stock liquidity as determinants of the momentum strategy of portfolio investment. Part 1. *World of Economics and Management (Vestnik NSU. Series: Social and Economic Sciences)*, 14(2):14–23. In Russian.

Tóth, B., Lempérière, Y., Deremble, C., de Lataillade, J., Kockelkoren, J., and Bouchaud, J.-P. (2011). Anomalous price impact and the critical nature of liquidity in financial markets. *Physical Review X*, 1(2):021006.

White, H. (2000). A reality check for data snooping. *Econometrica*, 68(5):1097–1126.

A Deviations from the pre-analysis plan

Implementation details not spelled out in the plan (decided before the factor results were seen). (i) When the Abdi–Ranaldo spread cannot be estimated for a share-month, its cost is set to the month's cross-sectional median cost (50 basis points if no estimate exists). (ii) Abnormal volume requires at least six listed months in the comparison window; all universes already require twelve months of trading, so the condition never binds. (iii) The CAPM-alpha variant runs the ladder on market-hedged returns $r_{k,t} - \hat{\beta}_k(r_{m,t} - r_{f,t})$ with $\hat{\beta}_k$ estimated once on the full sample; the bootstrap therefore ignores sampling error in $\hat{\beta}_k$. (iv) Daily returns across gaps of more than 14 calendar days (the 2022 suspension) are excluded from daily statistics for the market index as well as for shares. (v) The plan's characteristic table labels nine clusters, splitting downside risk from skewness and size from liquidity, while its description of the empirical-Bayes model specifies seven; the model uses seven, and the nine-label version gives the same replicated set (Section 5.4).

Changes made after the results were seen. (vi) Bug fix: the MCFTRR series used here has no print on 24 March 2022, so the daily market return on 25 March 2022 was first computed against the last pre-suspension print, contrary to the 14-day rule. It is now computed against the previous actual print and set to missing when the gap exceeds 14 days. The fix changes the characteristics that use daily market returns (mainly beta, idiosyncratic skewness and co-skewness) in a few months; it moves their baseline t-statistics by less than 0.05, and by at most 0.06 in any variant, and changes no discovery count in any specification. (vii) The power study uses 300 common outer resamples shared across the 25 factors and six Sharpe-ratio levels, with the inner Romano–Wolf bootstrap computed once per outer resample, instead of 100 independent outer replications per factor and level. Because the centred bootstrap statistics are invariant to the injected drift, this changes the precision of the simulation, not its estimand.

Exploratory analyses added after the results were seen. (viii) The Deflated Sharpe Ratio with a bootstrap null benchmark. (ix) Short-term returns within turnover terciles of UALL and turnover halves of U100 (Table 5). (x) The empirical-Bayes model without clusters and with nine clusters. (xi) Full ladders within each regime. (xii) Two parsimonious Fama–MacBeth specifications (Table 10). (xiii) The net-of-cost return of a short-term continuation strategy and the Romano–Wolf ladder on net returns. (xiv) The decomposition of the Romano–Wolf critical value in Section 5.2 (Gaussian and i.i.d. Newey–West benchmarks with and without the factors' correlation matrix, longer HAC lags, normal scores, rescaled bootstrap statistics, other seeds) and the simulation of family-wise error control in Appendix B. (xv) Details of the PBO splits.

Exploratory analyses added in the revision, after comments on the first version. (xvi) Bounds for unobserved delisting returns and a hold-to-resumption treatment of suspended shares (Section 5.5). (xvii) U100 rebuilt without the six depositary-receipt histories. (xviii) Residual momentum and idiosyncratic volatility with a market-plus-SMB model. (xix) Long- and short-leg returns, break-even borrow fees, the 2022 short-sale ban, short legs restricted to U50 or to index constituents, and an illustrative square-root price-impact charge (Section 5.7). (xx) Confidence intervals and minimum detectable changes for the regime differences, and the regime comparison for the average of the momentum survivors. (xxi) Correlations and spanning regressions among the survivors (Section 5.8). (xxii) The simulation of family-wise error rates was enlarged from 1,000 to 4,000 replications per design and extended to Holm and the $t > 3$ rule (the first version reported 6.2% and 6.5% for the Romano–Wolf rung), and a size-calibrated

Romano–Wolf level was added (Section 5.2). (xxiii) A discrepancy with the plan, found in the revision: the plan states that depositary receipts are not in the source and are therefore excluded, but six receipt histories are in the data and in the baseline universe (295 U100 stock-months from 2020). The baseline is kept as estimated; the universe that conforms to the plan, without these six issuers, gives the same five survivors (Table 6). The first version repeated the plan's statement and described each characteristic as defined as in its source; Section 3.2 and Table 12 correct both. (xxiv) Added in a second revision: four data-construction sensitivities that rebuild the panels from the raw data (a corrected dividend calendar; same-date payments treated as possible duplicates; the dates of the Yahoo-derived dividend source moved two sessions earlier or later; Appendix B), and a check of the volume-based characteristics for the shares that the price source takes from the T+2 board in 2013, implemented in a separate module, so that the configuration and data-construction modules hashed at the freeze are unchanged; look-ahead tests for universe membership and cost estimates and a continuity test of share counts through splits; and the index-restricted short leg of Section 5.7 is now evaluated only from January 2013, the first holding month with an index base in force (the first revision used the December 2012 base for earlier months). None of these changes alters a pre-specified result.

Apart from item (xxiii), no characteristic, sign, window, universe, breakpoint, weighting scheme or sample period differs from the plan.

B Data construction

Sources and provenance. Raw daily ISS share prices, the ticker dictionary including delisted securities, dividends, splits and MCFTRR are taken from the `moex-momentum` repository (Andreev, 2026) at commit `240a69d`; MOEX index calculation bases and the key-rate history from the `moexutils` repository (quantviews, 2026) at commit `9742060`; the current MOEX list of securities from the `moex-dataset` repository at commit `3320b44`. Prices in the first source are collected from every board on which a security traded, with the main post-2013 board (TQBR) taking precedence over the legacy boards, which take precedence over the odd-lot board. Dividends are collected from MOEX ISS and several secondary sources, reconciled, and checked for completeness against MOEX's export of register-closing dates.

Sessions. The session calendar is the union of the dates on which MCFTRR is published and the dates on which at least 20 shares traded (which adds official working Saturdays). Rows with non-positive closes or zero volume are treated as days without trades.

Dividends. A share's payments with the same record date are summed. The data contain 46 such groups (42 pairs and 4 triples); 40 of them, all pairs, belong to shares that were ever in U100. Many are a final and an interim dividend declared together, as NLMK and Severstal regularly do. I checked eight of the U100 pairs against issuer disclosures or press reports (NLMK 2015–2018, Severstal 2020, PIK and Inarctica 2021, Henderson 2024); all are distinct payments for different periods. Treating every group as a possible duplicate and keeping only its largest payment (an exploratory sensitivity, built on the corrected ex-date rule described below) drops 50 payments, 40 of them for shares that were ever in U100. Relative to the corrected rule alone it changes 33 U100 holding-month returns and no t-statistic by more than 0.04; relative to the baseline, 44 returns change, no t-statistic moves by more than 0.06, and the Romano–Wolf survivors are unchanged.

The plan's ex-date rule has two approximations. It locates the ex-date on the share's own trading days, so when a share does not trade around the record date the payout can be credited a few sessions early, in three U100 cases in the previous month. And it applies T0 settlement until September 2013, although from 25 March 2013 the source takes the prices of some of the largest shares, such as Sberbank and LUKOIL, from the T+2 board (TQBR); four U100 payouts are therefore placed two sessions late, within the same month. A corrected rule (exploratory) locates the ex-date on the exchange calendar, reads the settlement regime from the board on which the share traded, and credits the payout on the first own trading day on or after the ex-date. It changes 11 U100 holding-month returns, leaves the survivors unchanged and moves no baseline t-statistic by more than 0.03.

The Yahoo-derived source stores ex-dates whose convention is not uniform before September 2013. For Sberbank it gives 11 April 2012, the session before the record date of 12 April, and 12 April 2013, the session after the record date of 11 April; under the settlement regimes of the boards that supply the prices, the ex-dates were 13 April 2012 (T0) and 10 April 2013 (T+2). Both rules take the stored date as the ex-date. Moving every date of this source two sessions earlier (an exploratory sensitivity) changes 199 U100 holding-month returns, and moving every date two sessions later changes 197: most only slightly, through the price at which the payout is reinvested, and a few by moving a large payout into the adjacent month. In both cases the number of discoveries at every rung of the ladder and the survivors are unchanged, and the survivors' t-statistics move by at most 0.07; the largest change, 0.37, is for SEAS, whose baseline t-statistic is -0.41. Each of the four data sensitivities rebuilds the panels, characteristics and baseline portfolios from the raw data in separate folders and reruns the ladder (`scripts/09_run_data_sensitivities.sh`; definitions in `moexzoo/variants.py`).

Trading volume in 2013. The board switch noted above also affects volumes: from 25 March 2013 the source takes the daily rows of some large shares, volumes included, from the T+2 board, which carried a small part of their trading until the main board moved to T+2 on 2 September 2013. The recorded trading value of Sberbank, LUKOIL and VTB in April–August 2013 is 5 to 18% of its level in the first quarter, against 77 to 89% for Gazprom and Norilsk Nickel, which remain on the legacy board. In all, 23 shares have rows from a T+ board in that window. Their volume-based characteristics are understated in 2013, and abnormal volume, which compares a month with the preceding twelve, is distorted until August 2014. Setting the seven volume- and trading-based characteristics (abnormal volume, turnover, trading value, Amihud illiquidity, the variability of trading value, the zero-return proportion and the spread) of these shares to missing for formation months 2013:03–2014:08 (384 U100 stock-months; an exploratory sensitivity, `scripts/05h_board_volume.py`) lowers the t-statistic of abnormal volume from 3.52 to 3.38 and raises its Romano–Wolf adjusted p-value from 0.033 to 0.047; the number of discoveries at every rung and the survivors are unchanged, and no t-statistic moves by more than 0.45. Universe membership uses the median trading value over twelve months, which damps the effect: Sberbank, LUKOIL, VTB and RusHydro stay in U100 throughout.

Ticker changes. ISS records ticker changes (for example, TCSG to T); the predecessor's history is stitched into the successor's file, so the universe keeps only one of the two in any month. Redomiciliations create new securities with separate histories (for example, YNDX to YDEX). Where the predecessor was a depositary receipt with a different ticker (for example, FIVE to X5 or HHRU to HEAD), only the successor is in the data (Section 3.2).

Integrity tests. The replication package contains tests that (i) perturb prices, dividends, trading data and market returns after a formation month at four dates and verify that no characteristic changes; (ii) verify the continuity of adjusted prices around the 1:100 split of Norilsk Nickel (2024), the 5000:1 reverse split of VTB (2024) and the 1:100 split of Transneft preferred shares (2024); (iii) verify the placement and amount of a Sberbank dividend on its ex-date; (iv) verify the alignment of holding-period returns; (v) compare the Newey–West estimator and the Holm, Benjamini–Hochberg and Benjamini–Yekutieli procedures with `statsmodels`; (vi) check the Romano–Wolf implementation under a global null; (vii) perturb the trading data after a formation month (monthly trading value, trading days and last-trade dates; daily prices) and verify that universe membership and trading-cost estimates do not change; (viii) verify that adjusted share counts, and hence capitalisations, are continuous through all 14 splits that fall in the monthly panel, and that no adjusted share count changes by more than a factor of three from one month to the next (the continuity check is informative for the 6 splits with point-in-time index-base share counts on both sides; for the others the fallback count does not change across the split); and (ix) verify that the module of data-construction sensitivities reproduces the frozen adjustment exactly when given the plan's dividend rule. Tests (vii)–(ix) were added in the second revision. A separate simulation (4,000 replications in each of two designs, $B = 499$ inner resamples; exploratory) estimates the family-wise error rates of the Romano–Wolf rung, Holm and the $t > 3$ rule under the global null. For a multivariate-t process with five degrees of freedom, AR(1) coefficient 0.1 and the empirical correlation matrix of the factors they are 5.8%, 8.0% and 9.7%; for a stationary bootstrap of the demeaned factor returns, 7.3%, 9.8% and 11.5%, against a nominal 5% (Wilson 95% intervals in Section 5.2). The bootstrap test is mildly liberal at this sample length and the normal-theory procedures more so. Under the global null a family-wise error occurs exactly when the first-step bootstrap p-value of the largest $|t|$ is at or below the nominal level, so the simulated rate can be computed for any level. The calibrated level is the largest attainable level at which the simulated rate does not exceed 5%: 0.044 in the parametric design and 0.030 in the bootstrap design; the smaller is used.

C Additional results

Table 12: Implementation relative to the source definitions

ID	Source	Source definition	Implementation here
MOM12_1	Jegadeesh and Titman (1993)	Past 3–12-month returns; overlapping holding periods of 3–12 months	Months $t-11$ to $t-1$ (month t skipped)
MOM6_1	Jegadeesh and Titman (1993)	As above, six-month formation	Months $t-5$ to $t-1$
MOM12_7	Novy-Marx (2012)	Returns 12 to 7 months before the holding month	As in the source
RESMOM	Blitz et al. (2011)	Fama–French three-factor residuals, 36-month estimation; residuals over months -12 to -2 summed and scaled by their volatility	CAPM residuals, 35-month estimation (no book equity, hence no value factor); market-plus-SMB version in Table 6
HIGH52	George and Hwang (2004)	Price relative to the 52-week high; overlapping multi-month holding periods	As in the source, one-month holding
STR	Jegadeesh (1990)	Prior-month return	As in the source
LTR	De Bondt and Thaler (1985)	36-month formation and holding periods	Months $t-35$ to $t-12$
SEAS	Heston and Sadka (2008)	Same-calendar-month returns at annual lags, effect lasting up to 20 years	Mean over annual lags 1–5 (at least two)
BETA	Frazzini and Pedersen (2014)	Betas from five-year correlations of three-day returns and one-year volatilities, shrunk towards one; beta-neutral, rank-weighted, leveraged portfolio	Dimson (1979) beta over 252 sessions; low-minus-high quintile spread, not beta-neutral (its CAPM alpha is the closer test)
IVOL	Ang et al. (2006b)	S.d. of daily Fama–French three-factor residuals, previous month	CAPM residuals; market-plus-SMB version in Table 6
TVOL	Ang et al. (2006b)	S.d. of daily returns, previous month	As in the source
MAX	Bali et al. (2011)	Maximum daily return, previous month	As in the source
DBETA	Ang et al. (2006a)	Beta on days with below-mean market returns, one year of daily data; main evidence contemporaneous	Same estimator, used to predict next-month returns
SKEW	Amaya et al. (2015)	Weekly realised skewness from five-minute returns	Skewness of daily returns over 252 sessions (no intraday data)
ISKEW	Boyer et al. (2010)	Expected idiosyncratic skewness from a cross-sectional forecasting model	Historical skewness of daily CAPM residuals over 252 sessions
COSKEW	Harvey and Siddique (2000)	Standardised coskewness from 60 monthly returns	From 252 daily returns
SIZE	Banz (1981)	Market value of equity	Log capitalisation; share counts partly not point-in-time (PIT variant)
AMIHUD	Amihud (2002)	Mean daily $\lvert r\rvert$ per unit of dollar volume over a year	As in the source (rouble value, 252 sessions)
DVOL	Brennan et al. (1998)	Dollar trading volume	Log mean daily rouble value, 252 sessions
TURN	Datar et al. (1998)	Share turnover over recent months	Rouble value over mean capitalisation, 12 months
ZERO	Lesmond et al. (1999); Bekaert et al. (2007)	Proportion of zero-return days, a transaction-cost measure; priced in emerging markets in the second source	Proportion of no-trade or zero-return sessions, 252 sessions
SPREAD	Corwin and Schultz (2012); Amihud and Mendelson (1986)	High–low spread estimator; spread premium in the second source	Estimator as in the first source, month t
CVVAL	Chordia et al. (2001)	Coefficient of variation of monthly dollar volume over 36 months	Over 12 months
ABVOL	Gervais et al. (2001)	Unusually high volume over a day or a week; returns over the following month	Mean daily value in month t relative to months $t-12$ to $t-1$
DY	Litzenberger and Ramaswamy (1979)	Dividend yield in cross-sectional CAPM regressions	Trailing 12-month gross dividends over price, payers only

Month t is the formation month and $t+1$ the holding month. Whatever the holding rule of the source, every portfolio here is an equal-weighted quintile spread in U100 held for one month (Section 4); "as in the source" refers to the measure, not to the portfolio rule.

Table 13: t-statistics across specifications

Factor	Base	VW	U50	UALL	Terc.	Lag	Ord.	Price	Ex-22	PIT	Hedged
MOM12_1	**4.72**	**3.77**	**3.60**	3.58	**4.55**	**4.51**	**4.86**	**4.06**	**4.64**	**4.72**	**4.95**
MOM6_1	**5.33**	2.61	2.46	3.16	**4.32**	**5.32**	**5.31**	**4.47**	**5.32**	**5.33**	**5.59**
MOM12_7	3.24	2.88	2.78	2.47	**3.59**	3.12	**3.48**	2.82	3.14	3.24	**3.44**
RESMOM	**4.90**	**4.04**	**3.85**	**3.85**	**4.41**	**4.92**	**5.06**	**4.70**	**4.78**	**4.90**	**4.99**
HIGH52	**6.07**	**4.00**	**4.20**	2.61	**6.24**	**6.47**	**6.29**	**4.80**	**6.06**	**6.07**	**5.90**
STR	**−4.96**	−2.40	−3.27	0.29	**−4.32**	**−5.46**	**−4.95**	**−3.79**	**−4.97**	**−4.96**	**−4.73**
LTR	0.19	0.81	0.67	0.89	−0.10	0.52	−0.21	0.89	0.34	0.19	0.16
SEAS	−0.41	−0.61	−0.21	1.62	0.24	−0.96	−0.10	−0.92	−0.61	−0.41	−0.40
BETA	1.52	1.77	1.51	1.91	1.61	2.15	1.73	1.13	1.50	1.52	1.73
IVOL	1.85	0.47	1.13	0.25	1.77	1.93	1.93	1.53	1.87	1.85	2.16
TVOL	1.46	0.05	1.28	0.49	1.59	1.45	1.56	1.14	1.36	1.46	1.86
MAX	0.13	−0.95	0.45	0.40	0.96	−0.08	0.32	0.05	0.07	0.13	0.27
DBETA	−0.89	−1.65	−1.37	0.23	−0.61	−0.93	−1.29	−0.78	−0.77	−0.89	−1.04
SKEW	1.46	−0.22	0.84	−0.66	1.07	1.58	1.52	1.35	1.83	1.46	1.47
ISKEW	0.59	0.54	0.60	−0.76	0.69	0.56	0.81	0.60	0.80	0.59	0.62
COSKEW	0.21	−1.48	−0.27	1.99	0.48	0.41	−0.11	0.43	0.28	0.21	0.18
SIZE	0.41	0.77	−1.24	2.45	0.24	0.56	0.22	0.85	0.37	−0.25	0.40
AMIHUD	−0.39	1.42	−1.20	2.00	−0.81	−0.11	−0.12	0.25	−0.78	−0.39	−0.36
DVOL	0.70	1.58	−0.36	3.11	−0.10	1.22	0.85	1.36	0.37	0.70	0.81
TURN	1.52	1.25	0.94	1.12	1.74	1.82	1.61	1.46	1.29	0.35	1.93
ZERO	−0.70	−0.10	−0.69	2.47	−0.73	−0.31	−0.72	−0.47	−0.74	−0.70	−0.68
SPREAD	−1.74	0.34	−1.09	−0.11	−1.29	−1.97	−1.68	−1.05	−1.89	−1.74	−2.01
CVVAL	2.79	1.26	1.63	−0.44	2.05	2.39	2.55	2.14	**3.29**	2.79	2.79
ABVOL	**3.52**	**3.92**	2.49	1.21	**3.79**	3.09	**3.45**	2.37	**3.85**	**3.52**	**3.72**
DY	0.37	−0.61	−0.25	1.24	0.32	0.15	0.96	0.05	0.26	0.37	0.43

Newey–West t-statistics of the oriented long–short portfolios. Bold: rejection by the Romano–Wolf rung of that specification (either sign). Base: baseline; VW: capped value weights; U50, UALL: universes; Terc.: terciles; Lag: one-session implementation lag; Ord.: preferred shares dropped when the ordinary share is in U100; Price: price-only holding returns; Ex-22: excluding 2022:02 and 2022:03; PIT: point-in-time size and turnover; Hedged: market-hedged returns $r_{k,t} - \hat{\beta}_k(r_{m,t} - r_{f,t})$, whose t-statistic differs slightly from the regression $t(\alpha)$ in Table 3.

Table 14: Empirical-Bayes posteriors

			Clustered prior (pre-specified)				No clusters*	
Factor	Cluster	OLS mean	Post. mean	2.5%	97.5%	$P(\mu > 0)$	Post. mean	$P(\mu > 0)$
MOM12_1	Momentum	2.28	0.89	0.43	1.36	1.00	0.40	1.00
MOM6_1	Momentum	2.19	0.79	0.37	1.21	1.00	0.43	1.00
MOM12_7	Momentum	1.54	0.71	0.29	1.12	1.00	0.20	0.90
RESMOM	Momentum	1.64	0.82	0.41	1.23	1.00	0.47	1.00
HIGH52	Momentum	3.08	0.83	0.41	1.25	1.00	0.46	1.00
STR	Reversal	−1.96	−0.20	−0.57	0.17	0.15	0.04	0.60
LTR	Reversal	0.08	−0.14	−0.52	0.24	0.24	0.24	0.92
SEAS	Seasonality	−0.12	−0.00	−0.40	0.39	0.49	0.10	0.74
BETA	Low risk	0.69	−0.04	−0.36	0.28	0.40	0.04	0.59
IVOL	Low risk	0.85	0.03	−0.27	0.34	0.59	0.35	1.00
TVOL	Low risk	0.72	−0.01	−0.34	0.31	0.47	0.27	0.98
MAX	Low risk	0.06	−0.10	−0.43	0.22	0.26	0.08	0.72
DBETA	Skewness/downside	−0.35	0.22	−0.10	0.54	0.91	0.20	0.89
SKEW	Skewness/downside	0.56	0.30	−0.05	0.64	0.95	0.49	1.00
ISKEW	Skewness/downside	0.24	0.17	−0.13	0.48	0.87	0.24	0.96
COSKEW	Skewness/downside	0.06	0.21	−0.13	0.54	0.89	0.22	0.91
SIZE	Size/liquidity	0.18	0.44	0.19	0.69	1.00	0.44	1.00
AMIHUD	Size/liquidity	−0.16	0.40	0.14	0.65	1.00	0.28	0.97
DVOL	Size/liquidity	0.24	0.48	0.21	0.74	1.00	0.48	1.00
TURN	Size/liquidity	0.62	0.41	0.16	0.66	1.00	0.33	0.98
ZERO	Size/liquidity	−0.32	0.39	0.14	0.65	1.00	0.31	0.97
SPREAD	Size/liquidity	−0.83	0.36	0.11	0.62	1.00	0.15	0.83
CVVAL	Size/liquidity	0.98	0.39	0.16	0.63	1.00	0.40	0.99
ABVOL	Size/liquidity	1.40	0.46	0.20	0.71	1.00	0.38	0.99
DY	Payout	0.16	0.14	−0.32	0.59	0.72	0.25	0.93

OLS mean and posterior moments of the oriented monthly premium (%). Clustered prior: $\mu_k = \mu_0 + c_{j(k)} + w_k$ with the seven clusters of Table 2; hyperparameters by maximum marginal likelihood. No clusters: a single exchangeable prior (exploratory).

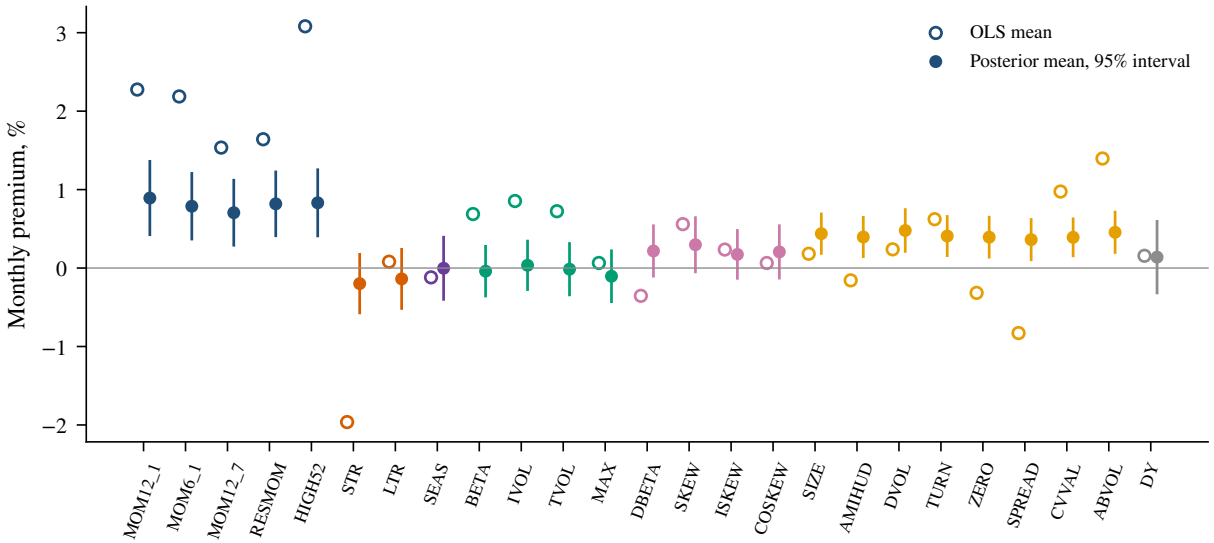


Figure 7: OLS means (open circles) and empirical-Bayes posterior means with 95% intervals (filled), clustered prior.

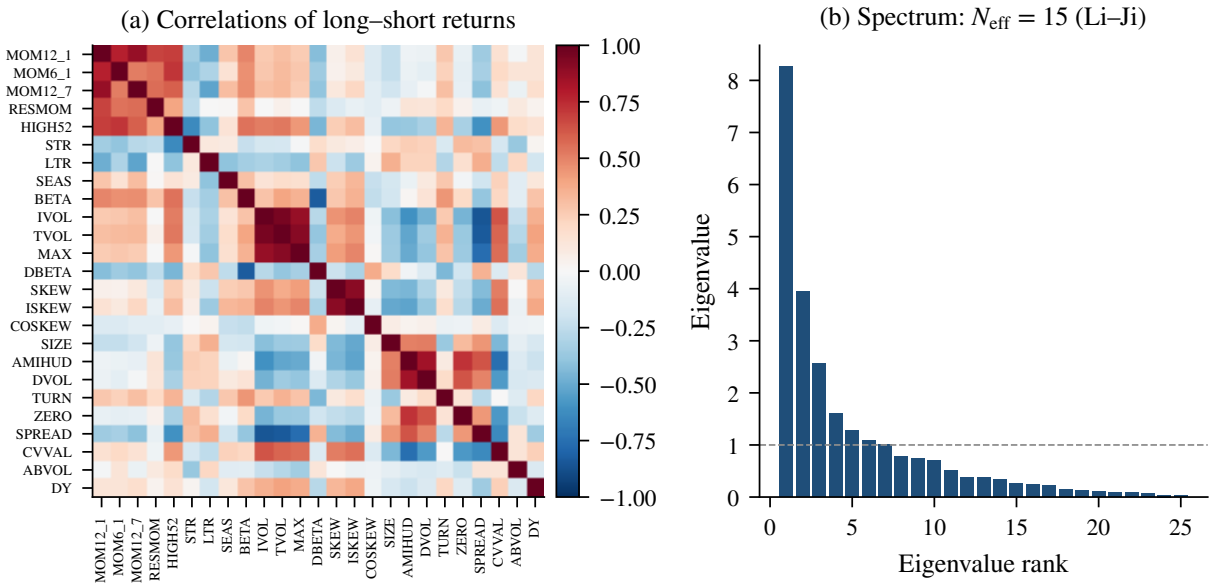


Figure 8: Correlations of the 25 oriented long–short returns (panel a) and the eigenvalues of their correlation matrix (panel b).